COMPLETE PASS-THROUGH IN LEVELS*

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Empirical studies find that the pass-through of input cost changes to prices is incomplete: a 10% increase in costs causes downstream prices to rise less than 10%, even at long horizons. Using microdata from gas stations, food products, and manufacturing industries, I find that incomplete pass-through in percentages often disguises *complete pass-through in levels*: a \$1/unit increase in input costs leads to \$1/unit higher downstream prices. Pass-through appears incomplete in percentages due to a gap between prices and costs. Complete pass-through in levels contrasts with workhorse macroeconomic models that feature homothetic industry demand systems. I identify an alternative class of demand systems that yields pass-through in levels and highlight four implications. First, measuring pass-through in percentages can lead to spurious evidence of asymmetry and size dependence. Second, pass-through in levels leads to systematic fluctuations in relative price and markup dispersion that are not associated with changes in allocative efficiency. Third, pass-through in levels can explain dynamics of industry gross margins, operating profits, and entry in the data that are at odds with workhorse models. Finally, incorporating pass-through in levels into an input-output model of the U.S. economy better matches the volatility of consumer price inflation and the response of inflation to identified shocks. *JEL codes:* E31, E32, L11.

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I. Introduction

Empirical work in macroeconomics and trade typically measures the pass-through of cost changes to prices in percentages. A large body of work studying pass-through in this way finds evidence of incomplete pass-through: when input costs rise by 10%, downstream prices increase by less than 10% (see Hellerstein 2008; Nakamura and Zerom 2010; De Loecker et al. 2016). Pass-through remains incomplete even at long horizons and after accounting for the input's share in variable costs. To account for this evidence, previous work has developed flexible demand systems in which the extent of pass-through is determined by forces such as market power, consumer heterogeneity, and the curvature of consumer preferences (e.g., Atkeson and Burstein 2008; Klenow and Willis 2016; Amiti, Itskhoki, and Konings 2019).

In this article, I instead measure the pass-through of input costs to downstream prices on an absolute, "dollars-and-cents" basis. I study a set of markets where I can precisely measure this pass-through in levels. Specifically, I study the pass-through of wholesale gasoline costs to retail stations' prices, the pass-through of food commodity costs to retail food prices, and the pass-through of input costs to output prices for industries spanning the U.S. manufacturing sector.

In nearly all cases, I find that firms exhibit complete pass-through in levels: a $1 per unit increase in input costs leads downstream prices to increase by $1. Complete pass-through in levels explains why pass-through measured in percentages appears incomplete: when price is greater than marginal cost, a $1 increase is a smaller percentage change in price than in marginal cost. Thus, complete pass-through in levels implies that the "log pass-through" is incomplete. I stress that my evidence of complete pass-through in levels applies to common cost shocks—that is, shocks that affect all producers in a market—and applies to shocks to the relative price of inputs, holding other prices in the economy (e.g., the aggregate price level) fixed.

Across the markets I study, I find that complete pass-through in levels explains not only the extent of incomplete log pass-through but also the cross-sectional variation in log pass-through across firms and products in a market. In response to a common cost shock, products with a larger gap between prices and input costs have lower log pass-through. These systematic differences in pass-through disappear when pass-through is instead measured in levels.

Complete pass-through in levels poses a challenge for workhorse models in macroeconomics that represent industry demand using homothetic demand systems. Homothetic industry demand systems are widely used because they impose that the quantity demanded from each firm depends only on its price relative to other firms and total industry sales, making them particularly tractable. However, I show that under standard assumptions about firms' profit maximization and conduct, homothetic demand systems predict that industry-wide cost shocks are passed through completely in logs rather than in levels. Complete log pass-through in turn implies that the pass-through in levels is equal to firms' gross markups and thus exceeds one.

This result arises because homothetic demand systems satisfy a property that I call *scale invariance*: a proportional change in all firms' prices leaves firms' demand elasticities unchanged. Thus, firms maintain constant percentage markups in response to an industry-wide cost shock. This applies to the standard constant elasticity of substitution (CES) preferences and to richer demand systems designed to account for incomplete pass-through, such as Kimball (1995) or nested CES preferences (Atkeson and Burstein 2008). While those richer demand systems can accommodate incomplete pass-through of idiosyncratic shocks that affect only a subset of firms in an industry, they uniformly predict complete log pass-through of common cost shocks, at odds with my evidence of pass-through in levels.

Thus, explaining complete pass-through in levels requires departing either from scale-invariant industry demand or from standard assumptions about firm conduct. I argue that one natural way of generating complete pass-through in levels is to use *shift-invariant* demand systems. When demand is shift invariant with respect to the prices of a set of firms, a uniform shift in those firms' prices scales each firm's residual demand curve by a constant factor. As a result, when firms experience a common cost shock, they retain fixed "additive markups"—defined as the absolute gap between price and marginal cost—and pass the cost shock through to prices one-for-one in levels.

Though the class of shift-invariant demand systems excludes homothetic preferences, it encompasses a variety of alternative models. For example, it includes the nested and mixed logit models, where heterogeneity in firm and consumer attributes generates rich patterns of substitution across firms (e.g., Nevo 2001). It also includes several models of spatial competition, such as the Hotelling (1929) and Salop (1979) models, in which firms' market power is derived from the transport costs that consumers incur to visit nearby versus faraway stores. Demand systems that satisfy shift invariance can also vary flexibly in their predictions for how firms pass through idiosyncratic cost shocks.

Shift-invariant demand systems generate complete pass-through in levels while maintaining standard assumptions about firms' profit maximization and imperfect competition. Of course, an alternative explanation for complete pass-through in levels is perfect competition. Under perfect competition, price equals marginal cost, and cost changes are reflected one-for-one in prices. Yet perfect competition is at odds with several other features of the data: sluggish price adjustment, price dispersion for identical products, finite firm-level demand elasticities, and evidence that firms' prices are elevated over available measures of costs. In other words, while the dynamics of prices relative to costs resemble perfect competition, price levels indicate some degree of imperfect competition. Shift-invariant demand systems offer a way of reconciling these seemingly conflicting empirical facts.[1]

I propose that complete pass-through in levels can explain several phenomena that are considerably more difficult to explain in standard models with homothetic preferences. I highlight four implications.

First, when pass-through is complete in levels, measuring pass-through in percentages can lead to asymmetry (different pass-through rates for cost increases versus decreases) and size-dependence (different pass-through rates for large versus small shocks). These patterns arise from imposing a log specification to measure pass-through. Measuring pass-through in percentages can also lead to systematic heterogeneity in pass-through by firm size and product quality, because if markups tend to increase with firm size or quality, then the log pass-through of common cost shocks will mechanically decrease with size and quality.

1. Alternatively, one can reconcile pass-through in levels with imperfect competition by relaxing assumptions about firm conduct, for example, through strategic interactions (e.g., Maskin and Tirole 1988), limit pricing, pricing rules-of-thumb (e.g., Hall and Hitch 1939), or managerial objectives beyond static profit maximization (e.g., Baumol 1959).

Second, pass-through in levels of common shocks leads to mechanical changes in the dispersion of relative prices and markups. When firms maintain fixed additive markups, an increase in input costs compresses the distribution of log prices and percentage markups. Indeed, I document across several industries that measures of relative price and markup dispersion tend to fall when input costs rise. Changes in price and markup dispersion are often interpreted as signs of changes in allocative efficiency, because when demand is homothetic, a decline in markup dispersion implies a reallocation of resources toward firms with higher initial markups and a reduction in misallocation. However, when I look at the response of quantities to common cost shocks in the data, even though relative price dispersion falls, I do not find evidence of the reallocations implied by homothetic demand.

Third, pass-through in levels can explain dynamics of industry gross margins, operating margins, and entry in the data that are at odds with workhorse models. In standard models of industry dynamics à la Dixit and Stiglitz (1977), homothetic preferences imply that an increase in input costs leads to higher profits per unit sold, which results in new firm entry or higher operating profits for incumbent firms. Neither of these patterns—swings in firm entry or operating profits—accompanies input cost fluctuations in the data. I show that pass-through in levels can bridge this gap between model and data. When firms exhibit complete pass-through in levels, a rise in input costs leads gross margins to fall, reducing the degree to which operating profits or entry need to adjust to maintain equilibrium. The response of gross margins to input cost changes across industries confirms these predictions.

Finally, I show that pass-through in levels can explain the low volatility of consumer price inflation relative to commodity prices. I calibrate an input-output model of the U.S. economy and compare two cases: one where firms maintain fixed percentage markups, as in standard models, and another where firms have additive markups, due to shift-invariant demand. Given the historical path of commodity prices, consumer price inflation in the model with fixed percentage markups is nearly twice as volatile as in the data. Although one could reduce this volatility by expanding the definition of firms' variable costs, doing so leads to implausibly low markups compared with empirical estimates. In contrast, the model with additive markups naturally matches the volatility of consumer price inflation in the data while allowing for markups consistent with the microeconomic evidence. The model with additive markups also better matches the response of consumer price inflation to identified cost shocks.

The outline of the article is as follows. Section II presents a simple example of pass-through in levels and logs. The next three sections present evidence of complete pass-through in levels: Section III measures pass-through in retail gasoline, Section IV in food products, and Section V in manufacturing industries. Section VI discusses implications for pass-through measurement, price and markup dispersion, and industry dynamics. Section VII characterizes shift- and scale-invariant demand systems. Section VIII applies these demand systems in an input–output model, and Section IX concludes.

I.A. Related Literature

This article relates to a large literature that studies theoretical and empirical determinants of pass-through.[2] I focus on the long-run pass-through of cost shocks that affect all firms in a market. Thus, I abstract from two topics that have generated large empirical literatures: (i) the pass-through of idiosyncratic shocks that only affect some firms in a market, and (ii) how rigidities influence the speed of transmission.

Whereas most studies in macroeconomics and trade measure pass-through on a percentage basis, there are several previous papers that measure pass-through in levels in specific contexts, especially in the industrial organization literature. I collect a list of previous studies that measure pass-through in logs and levels in Online Appendix Table A1. (Given the vast literature on pass-through, I focus on papers that study the pass-through of industry-wide cost shocks, rather than idiosyncratic shocks, and that measure pass-through using reduced-form methods, rather than simulating pass-through using a structural model.) The majority of studies in Online Appendix Table A1 are unable to reject complete pass-through in levels, and point estimates in many of the studies are tightly estimated around one. A few exceptions find evidence of under- or overshifting of excise tax changes, perhaps due to retailers rounding posted prices up or down to discrete

2. See Weyl and Fabinger (2013), Burstein and Gopinath (2014), Mrázová and Neary (2017), Miravete, Seim, and Thurk (2023, 2025), and the list of empirical studies in Online Appendix Table A1.

price points (see Conlon and Rao 2020). However, studies that use larger-sized tax changes or a larger sample of tax change events tend to find complete pass-through in levels, consistent with my results.

Two closely related studies in this literature are Nakamura and Zerom (2010) and Butters, Sacks, and Seo (2022). Nakamura and Zerom (2010) document that retail and wholesale coffee prices move one for one with coffee commodity prices in levels. Butters, Sacks, and Seo (2022) show that retailers exhibit complete pass-through in levels of various local cost shocks, including excise taxes, shipping costs, and wholesale prices. I document that complete pass-through in levels extends beyond retail and fast-moving consumer goods to a broader array of markets. Moreover, while both studies document that the average degree of markup adjustment coincides with pass-through in levels, I show that pass-through in levels in fact rationalizes the extent of pass-through observed across individual products, firms, and markets. Studies of gasoline markets also typically measure pass-through in levels rather than in logs (e.g., Karrenbrock 1991; Borenstein, Cameron, and Gilbert 1997; Deltas 2008) but do not explore why complete pass-through in levels is an appropriate benchmark.[3]

Finally, my characterization of shift- and scale-invariant demand systems relates to previous studies that explore the relationship between the shape of demand and pass-through. Bulow and Pfleiderer (1983) and Mrázová and Neary (2017) characterize how the elasticity and curvature of a firm's residual demand curve affect its pass-through of idiosyncratic cost shocks. My focus is instead on the pass-through of common cost shocks, shown in Section VII.E, and how the statistics that determine the pass-through of common and idiosyncratic shocks differ. Section VII relates the results to popular parameterizations of demand, such as the (scale-invariant) homothetic demand systems defined by Matsuyama and Ushchev (2017) and the (shift-invariant) class of linear random utility models defined by McFadden (1981) and Anderson, de Palma, and Thisse (1992).

3. For example, Borenstein (1991, 364) notes, "Though standard economic theory indicates that the percentage markup over marginal cost is the correct measure of market power, the industry literature and analysis focuses on the retail/wholesale margin measured in cents."

TABLE I
EXAMPLE OF PASS-THROUGH IN LOGS AND LEVELS

					Pass-through	
	Initial	Change	New	% Change	Logs	Levels
Commodity cost (c)	\$1	+\$0.20	\$1.20	+20		
Other variable costs (w)	\$1	—	\$1.00			
Total marginal cost	\$2	+\$0.20	\$2.20	+10		
Desired output price (p^*)						
Fixed percentage markup	\$4	+\$0.40	\$4.40	+10	1.0	2.0
Fixed additive markup	\$4	+\$0.20	\$4.20	+5	0.5	1.0

II. PASS-THROUGH IN LOGS AND LEVELS: A SIMPLE EXAMPLE

To begin, I illustrate the differences between complete pass-through in logs and levels in a simple example. Consider a firm that produces an output good using two inputs. I assume that the firm has a constant returns, Leontief production technology, so that the cost of producing y units of the output good is $C(y)$:[4]

$$C(y) = y(c + w),$$

where c is the price of the first input (the commodity), w is the price of the second input, and units of each input required to produce one unit of the output good are normalized to one. Table I shows an example in which $c = \$1$ and $w = \$1$.

In many models,[5] firms' desired prices p^* are equal to marginal cost times a fixed percentage markup, μ:

$$p^* = \mu(c + w). \tag{1}$$

In the example in Table I, the markup is $\mu = 2$, resulting in an output price of $2(\$1 + \$1) = \$4$.

How does an increase in the commodity price, Δc, affect the price set by the firm? Under the pricing rule in equation (1), the

4. Constant returns, Leontief production seems appropriate for the markets I study: for example, producing an ounce of ground coffee requires a fixed amount of coffee beans. In Online Appendix B.3, I consider how pass-through changes if I relax Leontief production, constant returns to scale, or uncorrelated other variable costs. Each requires knife-edge conditions to deliver complete pass-through in levels.

5. For example, monopolistic competition with CES demand. Even in many models of variable markups, firms retain fixed percentage markups in response to common cost shocks, as we will see in Section VII.

change in the firm's desired price is

$$\Delta p^* = \mu \Delta c.$$

The pass-through in levels of a commodity price change to the firm's desired price is equal to the markup μ. Typically, in markets with imperfect competition, $\mu > 1$, and so the pricing rule in equation (1) predicts that pass-through in levels is greater than one.

Table I shows the pass-through of a \$0.20 increase in the commodity price when the firm has a fixed percentage markup. Since a \$0.20 increase in the commodity price increases marginal costs by 10%, the output price also rises by 10%, or \$0.40. The pass-through in levels is equal to the markup, $\mu = 2$. The "log pass-through" is complete if measured with respect to the percent change in marginal cost ($\frac{10\%}{10\%} = 1$) or equal to the cost share of the commodity input if measured with respect to the percent change in the commodity cost ($\frac{10\%}{20\%} = 0.5$).

I contrast the pricing rule in equation (1) with an alternate pricing rule in which the firm's gap between output price and marginal cost, measured in dollars and cents, does not change in response to commodity cost changes:

$$p^* = c + w + m. \tag{2}$$

I refer to m as a fixed additive markup. Under equation (2), the firm's desired price instead rises one for one with the commodity cost: $\Delta p^* = \Delta c$. As shown in Table I, when the firm has a fixed additive markup, the percent change in the output price appears incomplete relative to the percent change in marginal cost (5% versus 10%). The percent change in the output price relative to the commodity price is also incomplete relative to the initial cost share of the commodity input ($\frac{5\%}{20\%} = 0.25$ versus 0.5). In other words, complete pass-through in levels appears as incomplete log pass-through.

I emphasize two features of how the empirical settings I study relate to the stylized example in Table I. First, the cost shocks I study are input price changes that affect all producers in a market. That is, my estimates capture the response of prices to industry-wide, or "common," cost shocks, rather than idiosyncratic cost shocks that only affect one or a subset of producers in a market. In Section VII, I discuss how firms that exhibit complete pass-through in levels of common cost shocks may exhibit different pass-through rates for idiosyncratic cost shocks.

Second, as in the example in Table I, I study how firms pass through an input cost shock holding fixed all other prices, such as overhead costs, entry costs, and the price level in the economy at large. In this sense, my estimates reflect the pass-through of a relative-price shock to inputs. I do not suggest that pass-through in levels of these relative-price shocks means that firms would adjust incompletely to a rise in the overall price level (e.g., if all nominal variables were to double).

III. Evidence from Retail Gasoline

Retail gasoline provides an ideal laboratory to study pass-through because there are rich data on firms' input costs and gasoline prices exhibit little rigidity. The main analysis in this section uses data on the universe of retail gas stations in Perth, Australia, though at the end of the section I show that retail gasoline markets in the United States, Canada, and South Korea all exhibit similar patterns.

This section documents four facts. First, the long-run pass-through in levels of wholesale costs to retail prices is statistically indistinguishable from one. Second, long-run log pass-through is incomplete even relative to the share of gasoline in stations' marginal costs. Third, there is little heterogeneity in pass-through in levels across stations, but substantial variation in log pass-through: stations with a larger gap between prices and costs have lower log pass-through. Fourth, complete pass-through in levels explains both the cross-sectional heterogeneity in log pass-through and the overall level of incomplete log pass-through.

III.A. Data on Station-Level Prices and Wholesale Costs

I use station-level gasoline price data from FuelWatch, a Western Australia government program that has monitored retail gasoline prices since January 2001. Since 2003, FuelWatch has also provided data on daily spot prices for wholesale gasoline, called the terminal gasoline price, across six terminals used by retail stations. Previous studies using these data include Wang (2009) and Byrne and de Roos (2017, 2019, 2022).

Following Byrne and de Roos (2019), I take the minimum terminal gasoline price offered by the six terminals each day as the input cost faced by retail gas stations. Online Appendix Figure A1 shows the weekly average terminal gasoline price and the retail unleaded petrol (ULP) price for a single gas station from 2001 to 2022. The retail price is slightly above, but closely tracks, the terminal gasoline price. The gap between retail and wholesale prices visibly increases in 2010. Byrne and de Roos (2019) document that retail gasoline margins in Perth increased starting in 2010 due to the emergence of tacit collusion across stations, a feature of the market that I exploit later in the analysis.

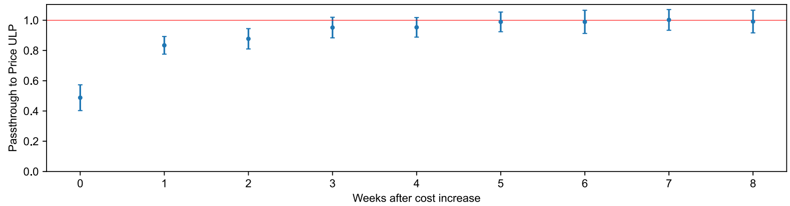
III.B. Empirical Results

1. Complete Pass-Through in Levels. I begin by measuring the pass-through in levels of wholesale gasoline costs to retail stations' prices. To measure the long-run pass-through of cost changes to prices, I estimate the distributed lag regression

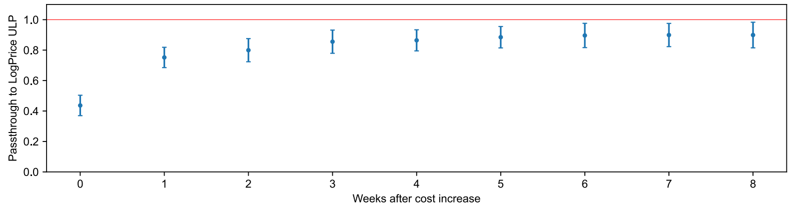
$$\Delta p_{it} = \sum_{k=0}^{K} b_k \Delta c_{t-k} + a_i + \varepsilon_{it}, \tag{3}$$

where Δp_{it} is the change in station i's retail price from week $t-1$ to t, Δc_{t-k} is the change in the input cost from $t-k-1$ to $t-k$, a_i are station fixed effects, and ε_{it} is a mean zero error term. The coefficients b_k measure the change in the output price associated with a change in input costs k periods ago. Accordingly, the long-run pass-through of a change in the input cost Δc to prices is given by the sum of the coefficients, $\sum_{k=0}^{K} b_k$. This specification is standard for measuring the long-run pass-through of cost changes to prices (e.g., Campa and Goldberg 2005, Nakamura and Zerom 2010), although I measure both price and cost changes in levels rather than in logs.

Our use of specification (3) is due to the fact that, as in Campa and Goldberg (2005) and Nakamura and Zerom (2010), my regressors are highly persistent. As I show in Online Appendix Table A2, autocorrelation coefficients for wholesale gasoline prices (and each of the other commodity price series I study) are very close to one, and I am unable to reject the hypothesis of a unit root in input prices. While commodity prices are approximately unit root, they appear stationary in first differences, enabling correct inference in specification (3). I also check in Online Appendix Table A3 that the direction of causality runs from upstream input costs to downstream prices and not vice versa, using Granger causality tests. In all cases, I do not find evidence that downstream prices Granger-cause upstream commodity prices.



(A) Pass-through in levels



(B) Pass-through in logs

FIGURE I

Pass-Through of Unleaded Gasoline Wholesale Costs to Prices in Levels and Logs

Panels A and B show cumulative pass-through estimated from specifications (3) and (4). Standard errors are two-way clustered by postcode and year, and standard errors for cumulative pass-through coefficients $\sum_{k=0}^{t} b_k$ and $\sum_{k=0}^{t} \beta_k$ are computed using the delta method.

Figure I shows the estimated pass-through of changes in ULP wholesale prices to station retail prices over a horizon of eight weeks. By three weeks, the pass-through in levels is statistically indistinguishable from 1, and the point estimate for long-run pass-through at eight weeks is 0.991 (standard error 0.038). Estimates of the pass-through of premium unleaded (PULP) wholesale prices to retail prices are similar (see Online Appendix Figure A2): the long-run pass-through in levels is 0.985 (0.036). Further increasing the horizon over which pass-through is estimated has little effect on the estimated long-run pass-through.

2. Incomplete Log Pass-Through. For comparison with previous studies that measure pass-through using percentage changes in input costs and prices, I estimate the long-run "log pass-through" using the specification,

$$\Delta \log p_{it} = \sum_{k=0}^{K} \beta_k \Delta \log c_{t-k} + \alpha_i + \varepsilon_{it}. \tag{4}$$

The long-run log pass-through is given by the sum of the coefficients, $\sum_{k=0}^{K} \beta_k$. The lower panel of Figure I shows that log pass-through of ULP costs to retail prices at eight weeks is 0.899 (0.043) and is statistically different from 1 at a 1% level. The log pass-through of PULP wholesale costs to retail prices is likewise significantly below 1 at 0.887 (0.041) (see Online Appendix Figure A2).

One reason that log pass-through may be incomplete is the presence of other variable costs besides gasoline. When stations have fixed percentage markups, the log pass-through of changes in the wholesale gasoline cost should equal the share of stations' marginal costs spent on gasoline. As previously documented by Byrne and de Roos (2019), retail prices in the Perth market follow weekly price cycles, jumping on Tuesdays or Thursdays and then falling over the course of the week. Under the assumption that gas stations never set prices below marginal cost,[6] I can use the days of the week at the lowest point of the price cycle to calculate an upper bound on the share of other variable costs in stations' marginal costs, and thus a lower bound for the cost share of gasoline. Averaging across all weeks and all stations, I find a lower bound for the cost share of gasoline of 0.97 for ULP and 0.96 for PULP. The estimated log pass-throughs, at 0.899 and 0.887, are significantly different from these cost shares at the 1% level. Thus, the log pass-through of gasoline costs is incomplete even after accounting for the cost share of gasoline.

3. Exploiting Variation in Markups. While the point estimates for pass-through in levels (0.991 and 0.985) are very close to 1, they do not rule out low markups that would be plausible in this setting. I further test for whether firms set fixed percentage markups by exploiting cross-sectional and time series variation in markups. If stations set fixed percentage markups, and if some stations have higher markups than others, then the pass-through in levels for high-markup stations should be higher than their low-markup counterparts. I estimate the specification,

$$\begin{aligned} \Delta p_{it} = \alpha + \delta \Delta c_t + \gamma \text{Avg. Markup}_{it} \\ + \beta(\Delta c_t \times \text{Avg. Markup}_{it}) + \varepsilon_{it}, \end{aligned} \tag{5}$$

6. This is the case in the Maskin and Tirole (1988) model of price cycles.

where Δp_{it} and Δc_t are changes in station i's price and the wholesale cost over the prior 16 weeks, Avg. Markup_{it} is a measure of station markups, and ε_{it} is a mean-zero error. The coefficient δ reflects the average rate at which wholesale cost changes are passed through to prices across gas stations, and the coefficient β estimates heterogeneity in this pass-through rate across stations with high versus low markups. I further allow stations with low- and high-markups to have differences in price trends that are independent of fluctuations in wholesale costs, which are estimated by the coefficient γ.

With fixed percentage markups, the coefficient on the interaction term $\beta > 0$. For example, if some stations set a fixed 2% markup and other stations set a fixed 5% markup, pass-through in levels should be 1.05 for the high-markup stations compared with 1.02 for the low-markup stations. On the other hand, if all stations exhibit complete pass-through in levels, the interaction coefficient $\beta \approx 0$. (An analogous intuition applies to time periods where stations charge higher or lower fixed percentage markups.)

I use two measures for Avg. Markup_{it}, along with instruments for both that are intended to isolate variation in markups from variation in non-gasoline variable costs. The first measure exploits variation in markups across stations: Avg. Station Markup_i is the average ratio of station i's retail price to the wholesale cost of gasoline over all weeks in the sample. To isolate variation in markups from non-gasoline variable costs, I instrument for Avg. Station Markup_i with the average amplitude of price cycles of station i, that is, the difference between the maximum and minimum retail margin charged by i in each week, averaged over all weeks. Although the ratio of stations' prices to wholesale costs may also capture variation in non-gasoline variable costs, this instrument isolates variation in markups across stations coming from the intensity of stations' price cycles.

The second measure instead exploits variation in markups over time: in each quarter t, I construct the average retail price over wholesale cost for all gas stations in Perth, denoted Avg. Quarter Markup_t. To instrument for Avg. Quarter Markup_t, I take advantage of the fact that the emergence of coordinated price cycles in the Perth market was, according to Byrne and de Roos (2019), "unrelated to market primitives." Online

TABLE II

COMPLETE PASS-THROUGH IN LEVELS: NO HETEROGENEITY BY STATION MARKUP

ΔPrice_{it}	OLS (1)	OLS (2)	IV1 (3)	OLS (4)	IV2 (5)
ΔCost_t	0.950	0.989[†]	0.952[†]	0.989[†]	0.971[†]
	(0.021)	(0.037)	(0.044)	(0.034)	(0.045)
ΔCost_t × Avg. Station Markup$_i$		−0.005	−0.000		
(net %)		(0.003)	(0.005)		
ΔCost_t × Avg. Quarter Markup$_t$				−0.004	−0.002
(net %)				(0.003)	(0.004)
N	312,215	312,215	312,215	312,215	312,215
R^2	0.89	0.89	0.89	0.89	0.89

Notes. The table reports the coefficients δ and β estimated using specification (5). Changes in retail prices and wholesale costs are taken over 16 weeks. For readability, I include Avg. Markup$_{it}$ on a net % basis (i.e., a markup of 1.1 is a 10% net markup). Column (3) (IV1) uses the average amplitude of stations' price cycles as an instrument for Avg. Station Markup$_i$. Column (5) (IV2) uses the quarterly R^2 of station margins on day-of-week dummies as an instrument for Avg. Quarter Markup$_t$. Standard errors two-way clustered by postcode and year are in parentheses. For estimates of the coefficient δ on ΔCost_t, [†] indicates estimates for which a pass-through of 1 is within the 90% confidence interval. For estimates of the coefficient β on the interaction term, * $p < .10$, ** $p < .05$.

Appendix Figure A3 shows that average gas station margins over time co-move closely with the degree of coordination in price cycles, measured as the R^2 from a regression of daily margins on day-of-week fixed effects. I use this measure of price coordination over time—the quarterly R^2 of station margins on day-of-week dummies—as an instrument for Avg. Quarter Markup$_t$.

Table II reports the results. Column (1) omits the average markup and interaction term. A $1 change in the wholesale cost of ULP over 16 weeks is associated with a $0.95 change in the retail station price over the same period. Columns (2)–(5) include the interaction of wholesale cost changes with markups, with columns (3) and (5) using the instruments discussed above. In all cases, the estimated coefficient on the interaction term $\beta \approx 0$, consistent with uniform pass-through in levels, rather than fixed percentage markups, across stations and time periods.

4. Pass-Through in Levels Explains Heterogeneity in Log Pass-Through. Table III reports estimates from an analogous specification that instead measures the pass-through of changes

TABLE III

INCOMPLETE LOG PASS-THROUGH IS EXPLAINED BY STATION MARKUPS

Δ log(Price)$_{it}$	OLS (1)	OLS (2)	IV1 (3)	OLS (4)	IV2 (5)
Δ log(Cost)$_t$	0.870	0.998†	0.968†	0.981†	0.970†
	(0.031)	(0.035)	(0.041)	(0.026)	(0.034)
Avg. Station Markup$_i$ × Δ log(Cost)$_t$ (net %)		-0.015^{**}	-0.011^{**}		
		(0.003)	(0.004)		
Avg. Quarter Markup$_t$ × Δ log(Cost)$_t$ (net %)				-0.010^{**}	-0.010^{**}
				(0.002)	(0.003)
N	312,215	312,215	312,215	312,215	312,215
R^2	0.88	0.89	0.89	0.89	0.89

Notes. The table reports the coefficients δ and β estimated using specification (6). Changes in log retail prices and log wholesale costs are taken over 16 weeks. For readability, I include Avg. Markup$_{it}$ on a net % basis (i.e., a markup of 1.1 is a 10% net markup). Column (3) (IV1) uses the average amplitude of stations' price cycles as an instrument for Avg. Station Markup$_i$. Column (5) (IV2) uses the quarterly R^2 of station margins on day-of-week dummies as an instrument for Avg. Quarter Markup$_t$. Standard errors two-way clustered by postcode and year are in parentheses. For estimates of the coefficient δ on Δ log(Cost)$_t$, † indicates estimates for which a pass-through of 1 is within the 90% confidence interval. For estimates of the coefficient β on the interaction term, * $p < .10$, ** $p < .05$.

in log costs to changes in log prices,[7]

$$\Delta \log p_{it} = \alpha + \delta \Delta \log c_t + \gamma \text{Avg. Markup}_{it} + \beta(\Delta \log c_t \times \text{Avg. Markup}_{it}) + \varepsilon_{it}. \tag{6}$$

Column (1) shows that a 1% change in wholesale costs over 16 weeks leads to a 0.87% change in retail prices, significantly below the cost share of gasoline. Columns (2)–(5) estimate specification (6), exploiting cross-sectional variation in markups (columns (2) and (3)) or time series variation in markups (columns (4) and (5)) in turn. Two findings emerge. First, higher markups lead to more incomplete log pass-through.[8] Second, the gap between price and costs appears to fully account for incomplete pass-through: the coefficient on $\Delta \log c_t$ in columns (3) and (5) shows that as net markups approach zero, the log pass-through is tightly estimated around the cost share of 0.97.

7. Since Table II suggests that stations' markups are "additive," it may be preferable to estimate specification (6) using a measure of stations' additive markups. I find that doing so yields similar results.

8. In fact, complete pass-through in levels predicts that the interaction coefficient in the log specification $\beta \approx -0.01$. If stations set prices $p = c + w + m$, where m is an additive markup, to a first order, $\Delta \log p \approx \chi \mu^{-1} \Delta \log c \approx \chi(1 - 0.01\mu^{\text{net},\%})\Delta \log c$, where $\chi = \frac{c}{c+w}$ is the cost share (0.96–0.97 in the data), $\mu = \frac{p}{c+w}$ is the percentage markup, and $\mu^{\text{net},\%} = 100(\mu - 1)$.

Thus, Table III shows that incomplete log pass-through is rationalized by the combination of complete pass-through in levels (documented in Table II) with a gap between stations' prices and marginal costs. Log pass-through is lower both for stations in the cross section and periods in the time series with higher markups. The size of the gap between output prices and gasoline input costs explains both the level of incomplete log pass-through and variation in log pass-through across stations.

5. Robustness. Online Appendix Table A4 compares pass-through estimates from Perth to estimates from retail gasoline markets in Canada, South Korea, and the United States (Online Appendix E describes the data sources for each). Incomplete log pass-through and complete pass-through in levels appear across all the studied markets. The evidence from other geographies suggests that complete pass-through in levels is not a quirk of the Australian data but describes price dynamics across a number of retail gasoline markets.

One might be concerned that my estimates of pass-through are biased downward due to reverse causality from downstream demand to commodity prices. While the Granger causality tests in Online Appendix Table A3 suggest that causality primarily runs from upstream commodity prices to downstream retail prices, as an additional check, Online Appendix Table A4 estimates pass-through using oil supply news shocks from Känzig (2021) to instrument for upstream cost changes. The instrumented regressions produce noisier estimates but remain qualitatively consistent with the baseline results.

IV. Evidence from Food Products

I explore the pass-through of commodity costs to retail prices for food products. I am able to measure pass-through in levels for these goods by carefully matching the amount of commodity inputs required to produce each downstream product.

For five out of six staple food products, I find that pass-through in levels is statistically indistinguishable from one. Using scanner data to compare individual products in product categories, I further document that uniform pass-through in levels explains the heterogeneity in log pass-through observed across products.

IV.A. Data on Food Retail and Commodity Prices

1. Retail Prices. For retail prices of food products, I use Average Price Data from the Bureau of Labor Statistics (BLS). In contrast to the consumer price index data, which reflect relative price changes, the Average Price Data track price levels for a select number of staple products. For each price series, the BLS chooses narrowly defined, homogeneous item categories (e.g., "Orange juice, frozen concentrate, 12 oz. can, per 16 oz.") to minimize input, quality, and package size differences among included items.

While the BLS Average Price Data allow me to study pass-through of commodity costs to retail prices over a long time series—many of the series record prices back to 1980—studying cross-sectional heterogeneity across products in a category requires richer data. For these investigations, I use NielsenIQ Retail Scanner data, which includes weekly barcode-level prices and quantities for products sold at participating stores from 2006 to 2020. These data are collected from point-of-sale systems in retail chains operating across the United States, reflecting over $2 billion in annual sales.

2. Commodity Costs. I match these retail food prices with data on commodity costs from the International Monetary Fund (IMF) Primary Commodities Prices database. These data draw from statistics of specialized trade organizations or from commodity futures markets—for example, the commodity price for frozen orange juice concentrate is from next-month futures contracts for the delivery of grade A frozen concentrated orange juice solids traded on the Intercontinental Exchange (ICE). Online Appendix Table A5 provides a full list of the commodity price series and the underlying data sources used by the IMF.

Measuring pass-through in levels requires carefully matching units from commodity prices to retail prices. For example, to measure pass-through of wheat commodity prices to retail flour prices requires knowing the quantity of wheat needed per pound of flour produced. To construct these mappings from commodity units to retail units, I rely on previous literature and on documentation from the U.S. Department of Agriculture. Online Appendix Table A6 provides the conversion factors from commodity prices to retail prices for each series and delineates the sources and assumptions used to build each conversion factor.[9]

3. Matched Products. Of the food products tracked by the BLS Average Price Data, six can be clearly matched to IMF commodity inputs. These are roasted ground coffee, sugar, ground beef, white rice, all-purpose flour, and frozen orange juice concentrate. Online Appendix Table A6 lists the corresponding Average Price Data series IDs. For some products, the Average Price Data include multiple price series that track different product sizes. I use all series available for each product when estimating pass-through. For three of these products—rice, flour, and coffee—I also investigate cross-sectional pass-through patterns by matching the food product to a NielsenIQ product category.[10]

IV.B. Empirical Results

1. Nearly All Products Exhibit Complete Pass-Through in Levels. I measure the pass-through of commodity costs to retail prices for each food product in levels and logs using the distributed lag specifications (3) and (4) described in Section III. As in the case of gasoline, each food commodity price series has an autocorrelation coefficient close to one but appears stationary in first differences (Online Appendix Table A2), enabling correct inference with the standard distributed lag specification. I also verify that retail price movements do not predict future changes in upstream commodity prices using Granger causality tests (Online Appendix Table A3) and by estimating the pass-through of leads

9. This careful matching of units is why measuring pass-through in levels is difficult for highly differentiated products. The challenges in measuring pass-through in levels, along with the fact that homothetic preferences imply a benchmark of complete log pass-through, are perhaps why pass-through in levels has not been measured across a wide set of markets previously. I have two exercises that show how one can estimate pass-through in levels even without this careful matching of units. First, in Online Appendix C.1, I exploit the fact that retailers set different prices for identical products to test for pass-through in levels at the retail level across a wider sample of NielsenIQ products. Second, Section V exploits my assumption of Leontief production to test for pass-through in levels in manufacturing industries.

10. The corresponding NielsenIQ product modules are “Rice - Packaged and bulk,” “Flour - All purpose - White wheat,” and “Ground and whole bean coffee.” Beef products are spread across several modules, and the “Sugar - granulated” and “Fruit juice - orange - frozen” modules have few unique products.

TABLE IV

LONG-RUN PASS-THROUGH OF COMMODITY COSTS TO RETAIL FOOD PRICES

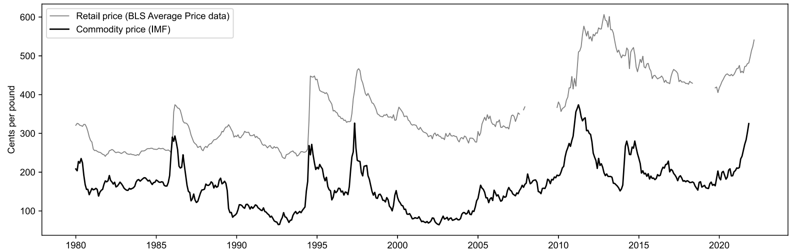
Commodity	Final good (BLS)	Pass-through (12 mos.) Logs	Pass-through (12 mos.) Levels
Arabica coffee	Coffee, 100%, ground roast	0.466	0.946†
		(0.051)	(0.099)
Sugar, no. 16	Sugar, white	0.370	0.691
		(0.035)	(0.072)
Beef	Ground beef, 100% beef	0.410	0.899†
		(0.068)	(0.126)
Rice, Thailand	Rice, white, long grain, uncooked	0.307	0.882†
		(0.049)	(0.169)
Wheat	Flour, white, all purpose	0.240	0.865†
		(0.048)	(0.160)
Frozen orange juice	Orange juice, frozen concentrate	0.327	0.974†
		(0.040)	(0.111)

Note: Long-run pass-through in levels and logs is $\sum_{k=0}^{K} b_k$ and $\sum_{k=0}^{K} \beta_k$ from specifications (3) and (4), using a horizon of $K = 12$ months. Robust standard errors in parentheses are cumulated using the delta method. † indicates estimates for which a pass-through of one is within the 90 percent confidence interval.

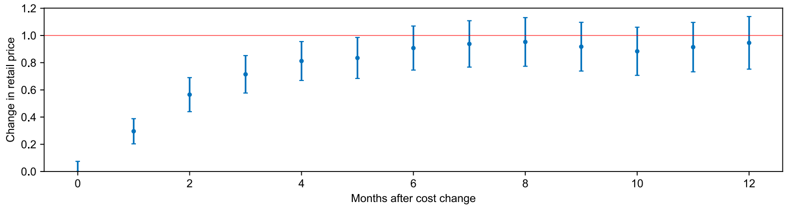
of commodity cost changes to retail prices (Online Appendix Table A7).

Table IV reports estimates of long-run pass-through in levels and logs from specifications (3) and (4) for six food products. In five of the products, long-run pass-through in levels is statistically indistinguishable from one. The exception is sugar, where the estimated pass-through in levels falls short of one.[11] Note that the pass-through in levels of commodity costs to retail prices is a particularly strict test of fixed percentage markups, because it should detect if any firm along the chain of producers from commodity to retailer sets a gross markup greater than one. Complete pass-through in levels implies log pass-through is incomplete and, as expected, I estimate that the long-run log pass-through is significantly below one for all six food products.

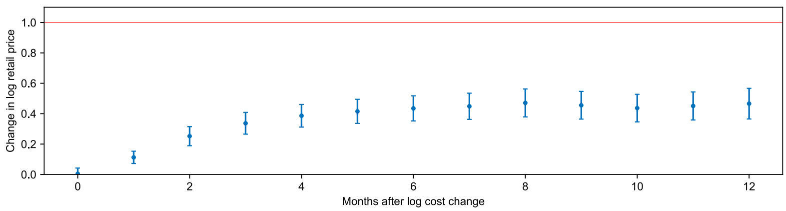
11. Online Appendix Figure A10 plots sugar commodity and retail prices extending back to 1960 from the U.S. Department of Agriculture. The pass-through in levels of sugar commodity prices to retail prices over the longer sample is 0.994 (std. err. 0.109). The lower estimate in the baseline sample is driven by incomplete pass-through of the rise and fall in sugar costs from 2009 to 2013. I speculate that incomplete pass-through over this period may have been due to offsetting movements in corn-based sweeteners, which are close substitutes.



(A) Arabica coffee commodity costs (IMF) and retail ground coffee prices (U.S. CPI)



(B) Pass-through in levels



(C) Pass-through in logs

FIGURE II

Pass-Through of Coffee Commodity Costs to Retail Prices

Panel A plots the time series of the commodity price from the IMF and the Average Price Data series from the BLS. The series are adjusted by the conversion factors in Online Appendix Table A6 so that the two series are in comparable units. Panels B and C plot the cumulative pass-through to month T, $\sum_{k=0}^{T} b_k$, from specifications (3) and (4), using a total horizon of $K = 12$ months.

Figure II shows an example of the price series and pass-through estimates for roasted ground coffee. As shown in Panel A, Arabica coffee commodity prices exhibit substantial volatility, with large spikes in 1986, 1994, 1997, 2011, and 2014 due largely to weather conditions in Brazil and Colombia. These run-ups in commodity prices are followed by increases in the retail prices recorded by the BLS. Panel B shows the pass-through in levels from coffee commodity prices to retail prices occurs with a lag but approaches complete pass-through by eight months and stays around one thereafter. The log pass-through, in Panel C, instead plateaus below one-half.[12] These results are consistent with Nakamura and Zerom (2010), who estimate pass-through in the roasted ground coffee market from 2000 to 2005. Analogous figures for the other five food products are in Online Appendix A.

2. Pass-Through in Levels Explains Variation in Log Pass-Through Across Products. The complete pass-through in levels documented in Table IV has predictions for price changes in the cross section of products. First, products that have higher markups and higher non-commodity input costs should exhibit lower log pass-through (as seen in the cross section of retail gas stations in Section III). Second, pass-through in levels should be similar across products regardless of their markups and non-commodity input costs.

To test these predictions, I use NielsenIQ data on rice, flour, and coffee products from 2006 to 2020. I define a product as a specific UPC (universal product code, or product barcode) sold at a specific retail chain, since prices for a UPC tend to be fairly uniform within retail chains (DellaVigna and Gentzkow 2019). In each quarter t, I calculate the price p_{it} of product i as the quantity-weighted average unit price over all transactions. For each product in each quarter, I then measure the change in the product's price over the next year in levels ($\Delta p_{it} = p_{it+4} - p_{i,t}$) and in logs ($\Delta \log p_{it} = \log p_{it+4} - \log p_{it}$). These price changes are measured year over year to avoid seasonality effects that may bias measures of price changes calculated over smaller time increments.[13]

Under the assumption that firms face identical commodity costs, I can use the unit price (e.g., the price per ounce of coffee) as a measure of each product's non-commodity variable costs and markups. Thus, to test the predictions for how pass-through in logs and levels varies with the level of non-commodity variable costs and markups, I group products in each product category by

12. Online Appendix Figure A4 shows that the pass-through in levels of coffee commodity cost changes to retail prices is similar using exchange rate shocks and weather shocks to instrument for commodity prices.

13. Nakamura and Steinsson (2012) point out that using product-level data to measure pass-through may bias measurement when there is frequent product turnover. For these categories, over 75% of products in each quarter are observed in the following year, and turnover does not appear correlated with commodity inflation in a way that would downward bias measured pass-through: the correlation of commodity inflation with turnover is −0.03 for rice, −0.09 for flour, and −0.09 for coffee products.

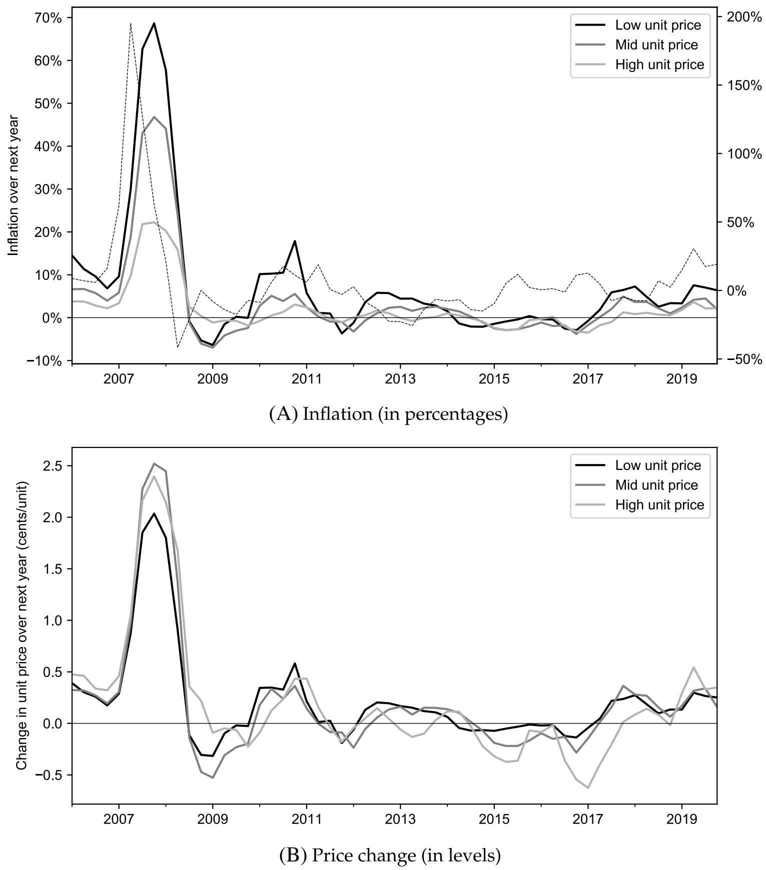


FIGURE III

Inflation and Price Changes of Rice Products by Tercile of Unit Price

Both panels plot price changes for rice products in the NielsenIQ scanner data. In each quarter, products are separated into three groups with equal quarterly sales by average unit price over the prior year. Panel A plots the sales-weighted average inflation rate over the next year for products in each group, alongside commodity rice inflation. Panel B plots the sales-weighted average change in price levels over the next year for products in each group

unit price in each quarter t. To ensure that these product groups capture persistent differences in unit price, I use products' average unit prices over the prior year.

As an example, Figure III plots average inflation rates and price changes in levels for these three groups of rice products. As shown in Panel A, a run-up in rice commodity prices into 2008 led to much higher inflation for rice products with lower unit prices—the average inflation rate for low–unit price rice products reached nearly 70% in 2008, compared with under 25% for high–unit price products.[14] These differences disappear when comparing the price changes in levels in Panel B: products in all unit price groups had roughly the same increase in absolute prices.

To formally test how pass-through in logs and levels varies in the cross section of products, I estimate the following specifications:

$$\Delta \log p_{it} = \alpha_i + \beta_1 \Delta \log c_t + \sum_{g=2}^{3} \beta_g \left(1\{G(i,t) = g\} \times \Delta \log c_t\right) + \varepsilon_{it}, \tag{7}$$

$$\Delta p_{it} = \alpha_i + \beta_1 \Delta c_t + \sum_{g=2}^{3} \beta_g \left(1\{G(i,t) = g\} \times \Delta c_t\right) + \varepsilon_{it}, \tag{8}$$

where $G(i,t) \in \{1, 2, 3\}$ is the unit price group of product i in quarter t, $\Delta \log c_t$ and Δc_t are changes in commodity prices over the next year in logs and levels, and α_i are product fixed effects. Panel A of Table V shows that the sensitivity of log retail prices to commodity inflation systematically declines with unit price across all three categories (rice, flour, and coffee). In contrast, Panel B finds few systematic differences in the sensitivity of retail price changes to commodity price changes in levels across unit price groups. Online Appendix Table A8 shows similar results if I instead split products into five unit price groups. Thus, evidence from all three categories suggests that products exhibit uniform pass-through in levels of commodity cost changes. The uniform pass-through in levels results in variable log pass-through rates across products, with lower log pass-through for higher-priced products.[15]

3. Pass-Through in Levels by Retailers. In Online Appendix C.1, I exploit the fact that different retail chains

14. The run-up in rice prices was prompted by adverse weather shocks to wheat-growing areas from 2006 to 2008, and subsequent trade restrictions by Vietnam, India, and other major rice-exporting countries to ensure adequate rice supply for their domestic markets. See Childs and Kiawu (2009) for a detailed account.

15. In follow-up work, Sangani (2025) shows that this pass-through behavior parsimoniously accounts for the bulk of variation in food-at-home inflation inequality over time.

TABLE V

HIGHER-PRICED PRODUCTS HAVE LOWER LOG PASS-THROUGH, BUT NO SYSTEMATIC DIFFERENCE IN PASS-THROUGH IN LEVELS

	Rice	Flour	Coffee
Panel A: In percentages	Δ Log retail price		
Δ Log commodity price × mid unit price	−0.075**	−0.007	−0.064**
	(0.014)	(0.009)	(0.015)
Δ Log commodity price × high unit price	−0.150**	−0.045**	−0.091**
	(0.022)	(0.009)	(0.017)
UPC fixed effects	Yes	Yes	Yes
N (thousands)	399.4	101.4	1,570.0
R^2	0.15	0.05	0.14
Panel B: In levels	Δ Retail price		
Δ Commodity price × mid unit price	0.059	0.027	−0.069
	(0.052)	(0.040)	(0.046)
Δ Commodity price × high unit price	0.042	−0.067	−0.099*
	(0.100)	(0.044)	(0.058)
UPC fixed effects	Yes	Yes	Yes
N (thousands)	399.4	101.4	1,570.0
R^2	0.07	0.05	0.14

Notes. Panel A reports results from specification (7), and Panel B reports results from specification (8). In each quarter, products are split into three groups with equal sales by average unit price over the past year; the mid– and high–unit price variables are indicators for the middle and highest-priced groups. Regressions are weighted by sales. Standard errors clustered by brand. * $p < .10$, ** $p < .05$.

often sell the same UPC at different prices (Kaplan and Menzio 2015) to test for pass-through in levels at the retail level for a broader sample of products. Under the assumption that retailers face identical wholesale costs, I show that the covariance between retailers' initial prices and their subsequent price changes for a UPC can be used to differentiate between complete pass-through in levels and log pass-through, even when wholesale costs are not directly observed. This allows me to test for pass-through in levels for a broader sample of nearly one million product UPCs in the NielsenIQ data. I find that (i) retailers exhibit complete pass-through in levels and (ii) uniform pass-through in levels explains variation in log pass-through across retailers.

V. EVIDENCE FROM MANUFACTURING INDUSTRIES

I extend the analysis to a panel of industries that span the U.S. manufacturing sector. I find that the pass-through of input cost changes to output prices across this broader sample of industries conforms with the predictions of pass-through in levels.

V.A. Data and Specification

I use the National Bureau of Economic Research NBER-CES Manufacturing Industry Database, which contains data on industry sales, costs, and input and output price indices for 459 four-digit SIC industries from 1958 to 2018. These data are constructed using the Census's Annual Survey of Manufacturers, the Census of Manufacturers, and the BLS Producer Price Index (PPI) program; I refer interested readers to documentation by Becker, Gray, and Marvakov (2021).

Unlike my earlier analysis of gasoline and food products, where I observed the level of input and output prices for specific products, the NBER-CES data include output and input price indices that reflect average percentage changes in prices. Nevertheless, under certain assumptions that I delineate below, I can estimate the pass-through in levels using changes in log price indices and the revenue-share of input expenditures.

Denote the pass-through in levels of a change in input costs to prices by $\rho^{\text{level}} \equiv \frac{\Delta p}{\Delta c}$. Rearranging yields

$$\frac{\Delta p}{p} = \rho^{\text{level}} \frac{cy}{py} \frac{\Delta c}{c},$$

where, on the right side, I multiply both the numerator and denominator by units of output y. Although I do not directly observe price levels or changes in price levels, to a first order, the ratio of the price change to the price level is equal to the change in the log price index, $\frac{\Delta p}{p} \approx \Delta \log p$. Thus, to a first-order approximation,

$$\Delta \log p \approx \rho^{\text{level}} \left(\frac{cy}{py} \right) \Delta \log c. \quad (9)$$

Under the assumption of constant returns, Leontief production, the term in parentheses $\left(\frac{cy}{py}\right)$ is simply the ratio of expenditures on the input to total sales.

Thus, I can test for pass-through in levels by estimating how changes in each industry's output price index relate to changes in its input price index multiplied by the revenue share of input costs. If firms exhibit complete pass-through in levels, changes in the output price index should move one for one with changes in the input price index multiplied by the revenue share of inputs. On the other hand, if firms have fixed percentage markups, the estimated pass-through in levels ρ^{level} will be greater than one.

V.B. Empirical Results

I estimate the specifications:

$$\Delta \log p_{it} = \rho^{\text{log}} \Delta \log c_{it} + \alpha_i + \varphi_t + \varepsilon_{it}, \tag{10}$$

$$\Delta \log p_{it} = \rho^{\text{level}} \left(\Delta \log c_{it} \times \left(\frac{\text{InputCosts}}{\text{Sales}} \right)_{it-1} \right)$$
$$+ \delta \Delta \log c_{it} + \gamma \left(\frac{\text{InputCosts}}{\text{Sales}} \right)_{it-1} + \tilde{\alpha}_i + \tilde{\varphi}_t + \varepsilon_{it}, \tag{11}$$

where $\Delta \log p_{it}$ ($\Delta \log c_{it}$) is the change in the log output (input) price index of industry i from year $t-1$ to t, $\left(\frac{\text{InputCosts}}{\text{Sales}} \right)_{it-1}$ is industry i's revenue-share of input expenditures in year $t-1$, and α_i ($\tilde{\alpha}_i$) and φ_t ($\tilde{\varphi}_t$) are industry and year fixed effects, respectively.

Specification (10) measures the reduced-form log pass-through of input price changes to output prices. Specification (11) implements equation (9). Assuming constant returns, Leontief production, equation (9) shows that the coefficient on the interaction term ρ^{level} in specification (11) identifies the pass-through in levels of the input cost change to output prices. Equation (9) also predicts that $\delta = \gamma = 0$, but I do not impose these restrictions for estimation. As discussed in Section II, I am interested in the pass-through of shocks to the relative price of a firm's inputs, so I measure changes in each industry's input and output price indices deflated by the consumer price index.[16]

Table VI presents the results from estimating specification (11) in the panel of manufacturing industries. In columns (1) and (2), I define input costs as each industry's materials costs. Column (1) shows industries exhibit incomplete log pass-through of materials input costs: a 1% increase in the materials input price index for an industry is associated with a 0.69% increase in output prices. Column (2) uses specification (11) to estimate the pass-through in levels. The estimated coefficient on the interaction term $\rho^{\text{level}} \approx 1$, consistent with complete pass-through in levels. Consistent with equation (9), I also estimate $\delta \approx 0$ and $\gamma \approx 0$.

Columns (3)–(6) find similar results when I extend the definition of inputs to include energy and production labor. In each case, I construct the change in the input price index as the weighted

16. Estimating equation (11) using nominal price changes does not meaningfully change the results in Table VI, but I show in Online Appendix C.2 that it can affect estimates of pass-through across different categories of inputs.

TABLE VI

PASS-THROUGH FOR MANUFACTURING INDUSTRIES

	Δ *Log Output Price$_t$*					
Inputs	Materials		+ Energy		+ Production labor	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log Input Price$_t$	0.690	0.079	0.704	0.005	0.796	0.052
	(0.072)	(0.132)	(0.073)	(0.134)	(0.083)	(0.232)
(InputCost/Sales)$_{t-1}$		0.004		0.008		0.023**
		(0.011)		(0.011)		(0.011)
Δ Log Input Price$_t$ $\times$ (InputCost/Sales)$_{t-1}$		0.947†		1.041†		0.984†
		(0.203)		(0.201)		(0.286)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	27,381	27,381	27,381	27,381	27,381	27,381
R^2	0.40	0.42	0.40	0.42	0.41	0.42

Notes. Columns (1) and (2) use input costs and prices for materials, columns (3) and (4) use input costs and prices for materials plus energy, and columns (5) and (6) use input costs and prices for materials, energy, and production labor. Input price inflation is an expenditure-weighted average across components of cost. Input and output price indices are deflated using CPI excluding food and energy. Standard errors (in parentheses) are two-way clustered by industry and year. For estimates of ρ^{log} and ρ^{level}, † indicates estimates for which a pass-through of one is within the 90% confidence interval. For estimates of the coefficients δ and γ, * $p < .10$, ** $p < .05$.

average of changes in price indices for each input, using the industry's expenditures on each input in the prior year as weights. I use the average hourly earnings of production and nonsupervisory employees in manufacturing as the price index for production labor across all industries to ensure that the labor price index is not biased by rent-sharing of profits with employees. In all cases, I find that the estimated coefficient on the interaction term ρ^{level} is very close to one, consistent with complete pass-through in levels of input cost changes to prices.

The derivation in equation (9) rests on several assumptions: homogeneous firms in each industry, constant returns to scale, and Leontief production technologies. Online Appendix B.3 characterizes the potential biases introduced by these assumptions. For example, decreasing returns to scale in production would bias up my estimates ρ^{level}—that is, would bias toward finding $\rho^{\text{level}} > \mu$ rather than $\rho^{\text{level}} = 1$. Allowing for substitutability between inputs (non-Leontief production) does not affect the first-order approximation in equation (9), but could bias downward the estimate of ρ^{level} for large cost shocks. The fact that I find $\rho^{\text{level}} \approx 1$ for various definitions of input costs in Table VI allays concerns that substitutability between inputs or correlated input prices biases the estimates of the pass-through in levels.

V.C. Robustness

To address concerns about reverse causality due to demand shocks from downstream industries, I also estimate specification (11) using an instrumental-variable approach. I use a decomposition of commodity price movements into demand shocks, industry productivity shocks, and commodity market shocks from Kabundi and Zahid (2023). By construction, the commodity market shocks they measure are orthogonal to aggregate demand shocks and downstream industry productivity shocks, so I use these commodity price shocks interacted with industry fixed effects as instruments for changes in each industry's input prices. Online Appendix Table A9 reports that the estimated coefficient ρ^{level} remains close to one using this approach.

A second concern is that nominal price rigidities may lead me to underestimate the long-run pass-through of input price changes to output prices. In Online Appendix Table A10, I estimate specification (11) using changes in input and output prices at horizons ranging from one to five years. If anything, the estimate of ρ^{level} slightly declines with the horizon, and I cannot reject the hypothesis that $\rho^{\text{level}} = 1$ for any horizon.

VI. Implications

I propose that pass-through in levels can be useful in understanding several features of the data. In this section, I discuss the implications of pass-through in levels for pass-through measurement, changes in the dispersion of relative prices and markups following cost shocks, and dynamics of industry margins and entry. Proofs for all propositions in this section are in Online Appendix B.1.

VI.A. Pass-Through Measurement

When firms exhibit complete pass-through in levels, Proposition 1 shows that using a log specification to measure pass-through can lead to asymmetry, size dependence, and systematic heterogeneity in pass-through across firms.

PROPOSITION 1 (ASYMMETRY, SIZE DEPENDENCE, AND HETEROGENEITY). Suppose firm j exhibits complete pass-through in levels, so that for a perturbation in firm j's costs dc_j, the firm's price changes by $dp_j = dc_j$. Let $\mu_j = \frac{p_j}{c_j}$ denote firm j's initial percentage markup. The measured "log pass-through" of firm j to a second-order approximation around $dc_j = 0$ is

$$\rho_j^{\log} = \frac{\Delta \log p_j}{\Delta \log c_j} \approx \frac{1}{\mu_j}\left[1 + \frac{1}{2}\frac{\mu_j - 1}{\mu_j}\Delta \log c_j\right].$$

Let $\rho_j^{\log}(\cdot)$ denote log pass-through as a function of the log cost change $\Delta \log c_j$. Then,

(i) log pass-through is asymmetric: $\rho_j^{\log}(x) > \rho_j^{\log}(-x)$ for any $x > 0$;
(ii) log pass-through is size-dependent: $\rho_j^{\log}(x) > \rho_j^{\log}(x')$ for any $x > x'$;
(iii) log pass-through is decreasing in markups: $\frac{\partial \rho_j^{\log}}{\partial \mu_j} < 0$ for small $\Delta \log c_j$.

A large literature documents asymmetries in log pass-through (see Peltzman 2000), and recent work finds that firms pass through large shocks at higher rates than small shocks (e.g., Gagliardone et al. 2025). Although these patterns could arise from genuine differences in how firms respond to cost changes, Proposition 1 shows that using a log specification naturally generates asymmetry and size dependence. With complete pass-through in levels, these effects arise because the log pass-through is not a stable statistic and depends on the size and direction of cost shocks.[17]

Measuring pass-through in logs can also lead to systematic patterns of heterogeneity. Previous work documents that log pass-through tends to decline with firm size (e.g., Berman, Martin, and Mayer 2012; Amiti, Itskhoki, and Konings 2019; Gupta 2020) and with product quality (Chen and Juvenal 2016; Auer, Chaney, and

17. In particular, complete pass-through in levels can explain asymmetry and size-dependence in the intensive margin of price adjustment—that is, the extent of pass-through conditional on a price change (e.g., Gagliardone et al. 2025, Figure 8), desired pass-through elicited in surveys (e.g., Bunn et al. 2024), or pass-through at long horizons (e.g., Peltzman 2000). It does not explain why the extensive margin of price adjustment, that is, the likelihood or frequency of price change, varies with the sign or size of cost shocks. Benzarti et al. (2020), Cavallo, Lippi, and Miyahara (2024), and Gagliardone et al. (2025) emphasize differences in the extensive margin of price adjustment in response to large versus small and positive versus negative shocks.

Sauré 2018). Complete pass-through in levels naturally generates both patterns if markups increase with firm size and product quality, as suggested by a large body of empirical work.[18]

VI.B. Changes in the Dispersion of Relative Prices and Markups

Pass-through in levels also implies that common cost shocks affect the distribution of firms' relative prices and markups. Intuitively, if firms retain fixed additive markups in response to common cost shocks, an increase in costs mechanically leads to a fall in the dispersion of relative prices and percentage markups, as I formalize in Proposition 2.

PROPOSITION 2 (DISPERSION IN RELATIVE PRICES AND REVENUE PRODUCTIVITY). Suppose firms $j = 1, ..., J$ have identical Leontief production technologies so that the cost of producing y units of output is $C(y) = (c + \beta w)y$, where c is the unit cost of the commodity, w is the wage rate, and β determines labor's weight in production, and each firm's price p_j is equal to marginal cost plus a fixed additive markup m_j. Then, an increase in the cost of the commodity input leads to a fall in

(i) the variance of relative prices $Var(\log p_j)$;
(ii) the variance of log revenue productivity $Var(\log \mathrm{TFPR}_j)$, where $\mathrm{TFPR}_j = \frac{p_j y_j}{C(y_j)}$; and
(iii) the variance of log labor revenue productivity $Var(\log \mathrm{LP}_j)$, where $\mathrm{LP}_j = \frac{p_j y_j}{w\beta y_j}$.

Consider first the effect of common cost shocks on the distribution of relative prices. Pass-through in levels predicts that increases in input costs should not affect the dispersion of prices in levels—the price distribution should simply shift to the right—but should reduce the dispersion of log prices. I test this prediction in Online Appendix Table A11 using the data on rice, flour, and coffee products explored in Section IV. As predicted by Proposition 2, I find that an increase in commodity prices has no significant effect on the dispersion of price levels but is associated with a decline in the dispersion of log prices across all three categories.

Likewise, complete pass-through in levels leads to a compression in the distribution of firms' percentage markups. Intuitively,

18. See Melitz (2018) for a survey of studies relating firm size and markups, and Atkin et al. (2015) and Sangani (2022) on product quality and markups.

when input costs rise, the firm-specific additive markups m_j become small relative to the common cost component, and so proportional differences across firms shrink. To test this prediction, I use measures of cross-sectional dispersion in total factor revenue productivity (TFPR) and labor revenue productivity (LP) in manufacturing industries from the BLS Dispersion in Productivity Statistics program. Foster, Haltiwanger, and Syverson (2008) and Hsieh and Klenow (2009) show that when firms have common technologies and face common input prices, differences in TFPR and LP reflect differences in firms' markups.[19] Consistent with Proposition 2, Online Appendix Table A12 finds that input price increases for manufacturing industries are associated with a decline in the dispersion of both TFPR and LP.

Declines in relative price and markup dispersion are often interpreted as indicating improvements in allocative efficiency. When industry demand is homothetic, firms' relative quantities are fully determined by their relative prices, so compressing the markup distribution reallocates resources toward firms with initially higher markups. This reallocation of resources to high-markup firms alleviates cross-sectional misallocation and raises efficiency.

If industry demand is not homothetic, a decline in markup dispersion need not reallocate resources to high-markup firms. Departing from homotheticity can therefore break the link between changes in markup dispersion and changes in allocative efficiency.[20]

I test whether the decline in markup dispersion induced by a common cost shock leads to the reallocations predicted by homothetic demand using quantity data for food products. A uni-

19. When firms have heterogeneous labor intensities, differences in TFPR and LP across firms reflect both differences in markups and technologies. Online Appendix B.1 extends Proposition 2 to the case where production labor intensity β_j varies across firms. When β_j is heterogeneous, the effect of commodity cost increases on $Var(\log \mathrm{TFPR}_j)$ and $Var(\log \mathrm{LP}_j)$ is theoretically ambiguous. However, I show in simulations that both variances tend to decrease with c even with substantial heterogeneity in β_j.

20. An important distinction is between the effect of marginal changes in markup dispersion and the effect of completely eliminating markup dispersion. Output is maximized when marginal revenue products are equalized across uses, regardless of whether demand is homothetic. But starting from an economy with dispersed markups, a marginal decline in markup dispersion need not improve allocative efficiency when demand is nonhomothetic. I provide an example with logit demand in Online Appendix B.1.

form absolute increase in products' prices represents a larger proportional increase for low-price products. If category demand is homothetic, this should trigger a reallocation of quantities away from low-price products and toward higher-price products. Online Appendix Table A13 tests how quantity shares of low- and high-priced rice, flour, and coffee products respond to changes in commodity prices. I find no effect on quantity shares of low- versus high-priced rice and flour products in response to commodity cost changes. For coffee, I observe if anything a reallocation toward low-price products when commodity costs rise, opposite the predictions of homothetic demand. Section VII discusses nonhomothetic demand systems that can both rationalize complete pass-through in levels and account for the absence of the reallocations predicted by homothetic demand.

Thus, complete pass-through of common cost shocks leads to systematic fluctuations in the dispersion of relative prices and percentage markups. Because real input costs for manufacturing industries tend to be procyclical, this mechanism can help account for the countercyclical TFPR dispersion documented by Kehrig (2011). Moreover, while input cost changes lead to declines in relative price and markup dispersion, they do not induce the quantity reallocations that would be predicted by homothetic demand and thus do not necessarily imply improvements in allocative efficiency.

VI.C. Dynamics of Industry Profits, Margins, and Entry

Whether firms maintain fixed percentage or additive markups in response to common cost shocks also has implications for the dynamics of industry profits, gross and operating margins, and entry. I show that the fixed additive markups consistent with pass-through in levels better match the response of these industry aggregates to input cost fluctuations.

The intuition for these results starts from the observation that if firms charge a fixed percentage markup over marginal cost, then as marginal costs increase, firms will make higher per unit profits. For example, gas stations charging a fixed 5% markup would make 5 cents per gallon sold when the wholesale cost of gasoline is \$1/gallon and 10 cents per gallon sold when the cost of gasoline increases to \$2/gallon. If aggregate industry demand is relatively inelastic, these higher per unit profits must show up as higher profits of existing firms or be dissipated through the entry of new firms. In other words, fixed percentage markups predict that an increase in input costs should lead to higher operating margins, new firm entry, or both—predictions that I show are counterfactual below. In contrast, if firms maintain fixed additive markups, a rise in input costs instead leads to an erosion of industry gross margins, a pattern that I observe across several industries.

1. Profits, Margins, and Entry in a Simple Industry Model. I consider a workhorse model of monopolistic competition, following Dixit and Stiglitz (1977) and Melitz (2003). An industry consists of a mass N of symmetric firms that produce varieties of an output good with a constant marginal cost c. I consider two cases: either (i) firms set price equal to marginal cost times a fixed percentage markup ($p = \mu c$), or (ii) firms set price equal to marginal cost plus a fixed additive markup ($p = c + m$).

In addition to variable costs of production, firms incur overhead costs f_o. I use π^{gross} to denote variable profits and $\pi^{\text{op}} = \pi^{\text{gross}} - f_o$ to denote operating profits net of overhead costs. I denote the output price chosen by firms in the symmetric equilibrium by p.

Aggregate industry demand is $Q = p^{-\theta}$. I assume that aggregate industry demand is inelastic ($\theta < 1$), which is the empirically relevant case for most of the industries studied in this article.[21] Firm symmetry allows me to express industry gross margins (i.e., gross profits as a percent of sales) and operating margins (i.e., operating profits as a percent of gross profits) as[22]

$$m^{\text{gross}} = \frac{\pi^{\text{gross}} N}{pQ}, \qquad \text{and} \qquad m^{\text{op}} = \frac{\pi^{\text{op}} N}{\pi^{\text{gross}} N}.$$

I close the model by specifying how the mass of firms evolves. Two common approaches are to assume a fixed mass of firms or to assume free entry. I choose a general condition that nests both as

21. Online Appendix Table A14 estimates $\theta \approx 0.33$ for retail coffee, using commodity prices, exchange rates, and weather shocks as instruments for retail prices. Studies that estimate demand for comparable industries find similar results: for example, Miller and Weinberg (2017) estimate $\theta \approx 0.60$–0.72 for beer; Backus, Conlon, and Sinkinson (2021) estimate $\theta \approx 0.34$–0.40 for ready-to-eat cereal; Conlon and Rao (2024) estimate $\theta \approx 0.50$–0.55 for spirits; and Grieco, Murry, and Yurukoglu (2024) estimate $\theta \approx 1.07$–1.44 for automobiles.

22. I define operating margins as the ratio of operating profits to gross profits, rather than sales. This ratio is sometimes instead referred to as the operating-profit conversion rate or operating efficiency.

TABLE VII
GROSS MARGINS, OPERATING MARGINS, AND MASS OF FIRMS

		Gross margins dm^{gross}	Operating margins dm^{op}	Mass of firms $d \log N$
Complete log pass-through				
$\zeta = 0$	(Fixed mass)	0	> 0	0
$\zeta \in (0, \infty)$		0	> 0	> 0
$\zeta \to \infty$	(Free entry)	0	0	> 0
Complete pass-through in levels				
$\zeta = 0$	(Fixed mass)	< 0	≤ 0	0
$\zeta \in (0, \infty)$		< 0	≤ 0	≤ 0
$\zeta \to \infty$	(Free entry)	< 0	0	≤ 0

special cases:

$$N = N_0(\pi^{\text{op}} - f_e)^{\zeta}, \tag{12}$$

where f_e is the entry cost and $\zeta \geqslant 0$ is the elasticity of the mass of firms to per firm profits. When $\zeta = 0$, the mass of firms is fixed at $N = N_0$. As ζ approaches infinity, there is free entry, and firms make zero profits net of the entry cost. Values of $\zeta \in (0, \infty)$ correspond to intermediate cases where entry responds to changes in operating profits, but not enough to keep operating profits in line with the entry cost.

Proposition 3 characterizes how industry aggregates—gross margins, operating margins, and the mass of firms—respond to changes in upstream costs.

PROPOSITION 3 (GROSS MARGINS, OPERATING MARGINS, AND ENTRY). Consider an increase in costs $dc > 0$. The response of industry gross margins, operating margins, and the mass of firms is shown in Table VII.

When firms have fixed percentage markups, an increase in costs leads to firms earnings higher profits on each unit sold. Because aggregate industry demand is inelastic, higher per unit profits lead to higher aggregate profits for incumbent firms or else are dissipated by the entry of new firms. In other words, fixed percentage markups imply that an increase in input costs must lead to higher operating margins, new firm entry, or both.

In contrast, when firms have fixed additive markups, per unit profits are invariant to changes in input costs. Thus, an increase in input costs no longer necessitates changes to operating profits or the mass of firms to maintain equilibrium. Instead, the industry equilibrium primarily adjusts to rising costs via a decline in gross margins.

2. Margins and Entry in the Data. I test the predictions of Proposition 3 using data on retail gas stations and manufacturing industries. For retail gas stations, I use data on gross and operating margins from the Census Annual Retail Trade Survey (ARTS) and from retail gas station sole proprietorships in the Internal Revenue Service Statistics of Income (SOI).[23] Data on the number of firms and establishments in each year comes from two sources: the Census Business Dynamics Statistics (BDS) and the Census Statistics of U.S. Businesses (SUSB).

Online Appendix Figure A11 shows the time series for retail gasoline gross margins, operating margins, and establishment growth rates. Gross margins exhibit a strong negative relationship with input prices (gross margins from the Census ARTS and IRS SOI have correlations with the wholesale gasoline price of −0.94 and −0.74, respectively). On the other hand, operating margins and firm entry appear largely unresponsive to input prices.

I test the relationship between outcome y_t and the input price c_t using the first-differences specification:

$$\Delta \log y_t = \alpha + \beta \Delta \log c_t + \varepsilon_t. \tag{13}$$

Table VIII, Panel A reports results from specification (13) using the price of wholesale gasoline as the measure of input costs and using gross margins, operating margins, and entry as outcome variables.[24] Neither operating margins nor entry significantly increase with input costs, as would be predicted by the model with fixed percentage markups. Instead, rising input prices lead to a significant decline in gross margins, as predicted by the model with fixed additive markups.

23. Sole proprietorships account for one-fifth of U.S. retail gas stations in 2016. I define sales as income from sales and operations in the SOI. Gross margins are gross profits (sales minus cost of sales) as a percent of sales. Operating margins are operating income as a percent of gross profits. To construct operating income, I start with reported net income, add back taxes and interest payments (payments of mortgage interest and other interest on debt), and subtract income from sources other than sales and operations.

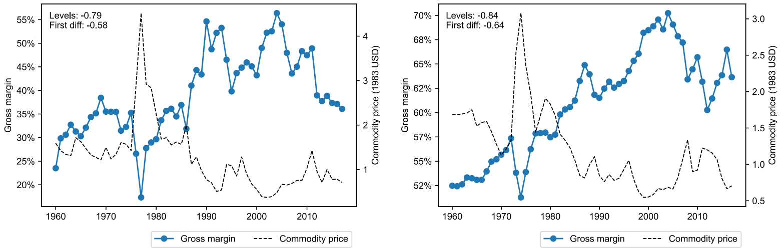
24. I obtain similar results using crude oil spot prices or nominal rather than deflated wholesale prices.

TABLE VIII

RESPONSE OF GROSS MARGINS, OPERATING MARGINS, AND ENTRY TO INPUT PRICE CHANGES

	Δ Log gross margin		Δ Log operating margin		Δ Log num. estabs.	
Source	ARTS (1)	IRS (2)	ARTS (3)	IRS (4)	BDS (5)	SUSB (6)
Panel A: Retail gasoline						
Δ Log Wholesale Price$_t$	−0.263**	−0.291**	0.490	0.125	−0.002	0.001
	(0.045)	(0.061)	(0.328)	(0.284)	(0.006)	(0.007)
N	39	26	15	26	39	24
R^2	0.54	0.49	0.20	0.01	0.00	0.00
Panel B: Manufacturing industries						
Δ Log Input Price$_t$	−0.188**	0.154	0.071	−0.086	0.007	−0.028
	(0.039)	(0.103)	(0.048)	(0.130)	(0.013)	(0.044)
Δ Log Input Price$_t$ × Inputs/Sales$_{t-1}$		−0.504**		0.270		0.049
		(0.188)		(0.221)		(0.063)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	27,381	27,381	27,305	27,305	18,201	18,201
R^2	0.05	0.11	0.02	0.04	0.22	0.23

Notes. Panel A presents results for retail gasoline. The wholesale gasoline price is from the EIA, deflated to constant dollars. ARTS is the Census Annual Retail Trade Survey, IRS are statistics for sole proprietorships, BDS is the Census Business Dynamics Statistics, and SUSB is the Census Statistics of U.S. Businesses. HAC-robust standard errors are in parentheses. Panel B reports results for manufacturing industries. The input price is the price index for materials, deflated to constant dollars; Inputs/Sales is the ratio of material costs to sales; gross margins are sales less material costs as a share of sales; operating margins are sales less material costs, energy costs, and payroll, as a share of gross profits; and num. estabs. are counts of establishments from the Census Business Dynamics Statistics (BDS). Standard errors in parentheses are two-way clustered by industry and year. ** $p < .05$.



(A) Roasted coffee manufacturing *vs.* coffee commodity price (B) Bread, cake & related products manufacturing *vs.* wheat commodity price

FIGURE IV

Gross Margins and Input Commodity Prices for Two Food Manufacturing Industries

Gross margins are sales minus costs of goods sold as a share of sales, from the NBER-CES manufacturing database. Annual coffee and wheat commodity prices from 1960–2018 are from the UN Trade and Development data hub, deflated using the CPI excluding food and energy

Similar patterns emerge for manufacturing industries. Figure IV plots industry gross margins against commodity costs for two manufacturing industries that use coffee and wheat as inputs. As predicted by the model with pass-through in levels, gross margins exhibit a strong negative correlation with input prices. Table VIII, Panel B reports how margins and firm entry respond to changes in input prices for the full set of manufacturing industries in the NBER-CES database. The response of all three industry aggregates to input costs is consistent with the predictions of additive markups in Proposition 3: gross margins exhibit a strong negative response to input price increases, whereas operating margins and entry show no significant response. Moreover, I find that the response of gross margins to input price changes scales with the revenue share of input costs, consistent with the predictions of additive markups.

Thus, complete pass-through in levels offers a way of reconciling the response of industry aggregates to cost fluctuations with simple models of industry dynamics. In particular, it offers an explanation for why industry gross margins fall when input costs rise—a pattern also observed in the exchange rate literature, where Hellerstein (2008) and Goldberg and Campa (2010) document that distribution margins as a percent of sales contract with increases in import prices. More broadly, these results clarify how industry equilibrium clears in response to cost shocks. Whereas standard models rely on entry and exit to maintain zero-profit conditions, the modified model with pass-through in levels achieves equilibrium with less volatility in the number of firms or operating profits.

VII. Scale- and Shift-Invariant Demand Systems

Many workhorse models in macroeconomics and trade use homothetic industry demand systems. I show that these demand systems are scale invariant and, under standard conduct assumptions, imply complete log pass-through of common cost shocks, at odds with my evidence. Explaining complete pass-through in levels therefore requires departing from scale-invariant demand or standard assumptions about firm conduct. I identify a class of shift-invariant demand systems that, under the same conduct assumptions, instead generate complete pass-through in levels of common cost shocks.

VII.A. Environment

Suppose there is a set of goods that is partitioned into a set of $J \geqslant 1$ inside goods and $K \geqslant 0$ outside goods. Denote the vector of prices for inside goods by $\boldsymbol{p} = (p_1, ..., p_J)$ and the vector of prices for outside goods by $\boldsymbol{p_0}$. The demand system $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ describes the quantity consumed of each good as a function of the prices of inside goods $\boldsymbol{p}$, the prices of outside goods $\boldsymbol{p_0}$, and income Y.[25]

Each inside good is produced and sold by a single firm. For $j \in \{1, ..., J\}$, firm j possesses a constant returns to scale production function with exogenous marginal cost c_j. I assume that each firm sets its price p_j to maximize profits, taking the prices set by other firms and consumer demand curves as given.

Assumption 1 (Nash in Prices). For each $j \in \{1, ..., J\}$, the price p_j is set to maximize firm j's profits, taking all other prices, income Y, and the demand system D as given.

While Nash in prices is a standard assumption in macroeconomic models with imperfect competition, this assumption may be

25. I assume throughout that D is a standard Marshallian demand system, and thus is homogeneous of degree zero in prices and income, that is, $D(\lambda \boldsymbol{p}, \lambda \boldsymbol{p_0}, \lambda Y) = D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$. A consequence is that all real variables—for example, quantities and percentage markups—are neutral with respect to changes in the aggregate price level.

violated because of collusion (as in the case of Perth gas stations after 2010), strategic interactions between oligopolistic firms, or Cournot competition, among other reasons. My goal in adopting Nash in prices is not to deny the possibility of other types of conduct but to identify conditions under which standard models do (or do not) reproduce our results on complete pass-through in levels.

Given an exogenous vector of outside prices $\boldsymbol{p_0}$, income Y, and vector of marginal costs $\boldsymbol{c} = (c_1, ..., c_J)$, an equilibrium is a vector of prices $\boldsymbol{p}$ and quantities $\boldsymbol{q}$ such that, for $j = 1, ..., J$, $q_j = D_j(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ and p_j maximizes the profits of firm j taking all other prices as given.

I am interested in the pass-through of a change in the costs of producing the inside goods, starting from an initial equilibrium. For all results that follow, I impose Assumption 2, which guarantees that such an equilibrium exists and that firm's residual demand curves are downward-sloping in this initial equilibrium.

ASSUMPTION 2 (EQUILIBRIUM EXISTENCE AND DOWNWARD-SLOPING DEMAND). Given outside prices $\boldsymbol{p_0}$, income Y, and marginal costs $\boldsymbol{c}$, (i) an equilibrium exists, and (ii) the own-price elasticity of demand for each inside good $\frac{\partial \log D_j}{\partial \log p_j}$ is finite, continuous, and strictly negative.

The assumption of finite elasticities of demand precludes perfect competition, in which firms face perfectly elastic residual demand curves. Of course, under perfect competition, firms' prices equal marginal costs and thus changes in marginal cost are passed through one for one in levels to prices. However, several other features of the markets that I study are at odds with perfect competition: these industries exhibit price dispersion for identical products, prices that are elevated over available measures of costs, and small and finite demand elasticities. Thus, I seek to characterize restrictions on demand that yield pass-through in levels in such environments with imperfect competition.

VII.B. Scale Invariance, Shift Invariance, and Pass-Through

I define the following properties of the demand system.

DEFINITION 1 (SCALE INVARIANCE). $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is scale invariant in $\boldsymbol{p}$ if there exist functions $\varphi_1, ..., \varphi_J$ such that $\frac{\partial \varphi_j}{\partial p_j} = 0$ for all j and, for any positive constant λ,

$$D_j(\lambda \boldsymbol{p}, \boldsymbol{p_0}, Y) = \lambda^{\varphi_j(\boldsymbol{p}, \boldsymbol{p_0}, Y, \lambda)} D_j(\boldsymbol{p}, \boldsymbol{p_0}, Y) \qquad \text{for all } j \in \{1, ..., J\}.$$

DEFINITION 2 (SHIFT INVARIANCE). $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is shift invariant in $\boldsymbol{p}$ if there exist functions $\psi_1, ..., \psi_J$ such that $\frac{\partial \psi_j}{\partial p_j} = 0$ for all j and, for any constant λ,

$$D_j(\boldsymbol{p} + \lambda \mathbf{1}, \boldsymbol{p_0}, Y) = \left(1 + \lambda \psi_j(\boldsymbol{p}, \boldsymbol{p_0}, Y, \lambda)\right) D_j(\boldsymbol{p}, \boldsymbol{p_0}, Y) \quad \text{for all } j \in \{1, ..., J\},$$

where $\mathbf{1} = (1, ..., 1)$ is a J-length vector of ones.

Scale and shift invariance are restrictions on how changes to the prices of all inside goods affect demand. Scale invariance imposes that if the price of every inside good is multiplied by the same factor, the quantity demanded of each good changes by a factor that does not depend on the good's price. Thus, a proportional price change to all inside goods leaves the elasticities of the residual demand curves facing each firm unchanged. Shift invariance instead restricts how the quantity demanded of each good changes when the price of each inside good changes by the same absolute amount. Under shift invariance, a uniform shift in all prices scales the demand schedule facing each firm, so that the level and slope of the residual demand curve for each firm scale by the same (potentially firm-specific) factor.

The main results in Propositions 4 and 5 show how these properties of demand determine the pass-through of common cost shocks to prices.

PROPOSITION 4 (COMPLETE LOG PASS-THROUGH). Consider a shock that increases the marginal cost of production for all inside goods proportionally by $d \log c_j = d \log c$ for all $j \in \{1, ..., J\}$, holding fixed outside prices $\boldsymbol{p_0}$ and income Y. If demand is scale invariant in $\boldsymbol{p}$, then each inside good's price increases by $d \log p_j = d \log c$.

When demand is scale invariant in inside prices, firms exhibit complete log pass-through of a common (proportional) cost shock. Intuitively, scale invariance implies that when all firms completely pass through the aggregate cost shock in logs, the elasticities of demand facing firms are unchanged, and firms have no further motive to change their percentage markups. Thus, firms exhibit fixed percentage markups when faced by common, proportional cost shocks.

Analogously, a common (absolute) cost shock to firms is passed through completely in levels when demand is shift invariant.

PROPOSITION 5 (COMPLETE PASS-THROUGH IN LEVELS). Consider a shock that increases the marginal cost of production for all inside goods by $dc_j = dc$ for all $j \in \{1, ..., J\}$, holding fixed outside prices $\boldsymbol{p_0}$ and income Y. If demand is shift invariant in $\boldsymbol{p}$, then each inside good's price increases by $dp_j = dc$.

When demand is shift invariant in inside prices, a uniform absolute increase to inside goods' prices scales the demand schedule facing each firm. This scaling means that each firm's desired additive markup, which is equal to the ratio of a firm's demand to the slope of its residual demand curve, remains unchanged. Thus, firms retain fixed additive markups in response to a common cost shock.

For the purpose of characterizing pass-through, I considered two different types of cost shocks: the first type of cost shock in Proposition 4 increased all firms' costs by a fixed proportion, and the second type of cost shock in Proposition 5 increased all firms' costs by a fixed amount. One may wonder whether the differences in pass-through behavior are attributable to differences in the type of aggregate cost shock. This is not the case: Proposition 6 shows that a demand system cannot be simultaneously scale invariant and shift invariant with respect to the same set of prices.

PROPOSITION 6 (SCALE AND SHIFT INVARIANCE ARE DISJOINT). If $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is scale invariant in $\boldsymbol{p}$, then $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is not shift invariant in $\boldsymbol{p}$. Equivalently, if $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is shift invariant in $\boldsymbol{p}$, then $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is not scale invariant in $\boldsymbol{p}$.

Proposition 6 shows that if a set of firms exhibit complete log pass-through of a common, proportional shock to costs, those firms cannot exhibit complete pass-through in levels of a common, absolute cost increase. Likewise, if a demand system implies complete pass-through in levels of a common absolute cost shock to a set of firms, those firms will not exhibit complete log pass-through of a proportional cost shock.

VII.C. Examples of Scale-Invariant Demand

Together, Propositions 4 and 6 imply that models with scale-invariant demand are inconsistent with pass-through in levels of common cost shocks (given Assumptions 1 and 2). Below, I show that homothetic demand systems commonly used in macroeconomics and trade fall into this category. Proofs for the examples that follow are in Online Appendix B.2.

EXAMPLE 1 (HOMOTHETIC PREFERENCES). Suppose there are J goods, and a representative consumer chooses consumption of each good q_j to maximize utility $U(q_1, ..., q_J)$ subject to the budget constraint $\sum_j p_j q_j = Y$. If U is homothetic, then the quantities $\boldsymbol{q}$ chosen by the consumer can be represented by a scale-invariant demand system $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$, where $\boldsymbol{p_0} = \emptyset$ and $\varphi_j = -1$.

Since homothetic demand systems are scale invariant, they predict that firms retain constant percentage markups in response to aggregate, proportional cost shocks. This result applies to any homothetic demand system, regardless of whether the demand system implies fixed percentage markups—as in CES demand—or whether it accommodates variable markups, as in Kimball (1995) or homothetic single-aggregator (HSA) (Matsuyama and Ushchev 2017) preferences. This result also applies regardless of whether firms are atomistic, as in models of monopolistic competition, or granular, as in the Atkeson and Burstein (2008) oligopoly model. Although the latter models can account for incomplete log pass-through of idiosyncratic cost shocks, they uniformly predict complete log pass-through of aggregate (proportional) cost shocks because of their scale invariance. Moreover, Proposition 6 implies that since homothetic demand systems are scale invariant, they are not shift invariant and thus are inconsistent with the evidence of complete pass-through in levels of common cost shocks.

An analogous result applies to the pass-through of industry-wide cost shocks in models where preferences are a CES nest over homothetic industry aggregates.

EXAMPLE 2 (NESTED HOMOTHETIC PREFERENCES). Suppose there is a continuum of industries indexed by $n \in [0, 1]$, each of which consists of J firms. A representative consumer maximizes

$$U = \left(\int_0^1 Q_n^{\frac{\sigma-1}{\sigma}} dn \right)^{\frac{\sigma}{\sigma-1}} \qquad \text{s.t.} \qquad \int_0^1 \sum_{j=1}^{J} p_{nj} q_{nj} \, dn = Y,$$

where p_{nj} is the price of firm j in industry n, q_{nj} is the quantity purchased from firm j in industry n, σ is the elasticity of substitution across industries, Y is the consumer's income, and $Q_n = u(q_{n1}, ..., q_{nJ})$ is a homothetic aggregate of consumption from firms in industry n.

Then, for any industry n, the quantities $\boldsymbol{q_n} = (q_{n1}, ..., q_{nJ})$ can be represented by a demand system $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$, where $\boldsymbol{p} = (p_{n1}, ..., p_{nJ})$ is the vector of prices for the J firms in industry n, $\boldsymbol{p_0}$ is the vector of prices of all firms outside industry n, and $D(\boldsymbol{p}, \boldsymbol{p_0}, Y)$ is scale invariant in $\boldsymbol{p}$ with $\varphi_j = -\sigma$.

Example 2 includes the nested CES demand system from Atkeson and Burstein (2008) and the nested Kimball demand system from Amiti, Itskhoki, and Konings (2019) as special cases. In these models, demand is scale invariant with respect to the prices of firms in an industry, so firms exhibit complete log pass-through of industry-wide (proportional) cost shocks. Thus, any model of industry demand that can be expressed as a special case of Example 2 will not exhibit complete pass-through in levels of common cost shocks.

VII.D. Examples of Shift-Invariant Demand

Clearly, complete pass-through in levels requires deviating from either Assumptions 1 and 2 or from scale-invariant demand. Given the evidence that quantities do not respond to common cost shocks as homothetic demand would predict, a natural choice is to adopt demand systems that generate complete pass-through in levels while retaining standard conduct assumptions.

I present examples of such shift-invariant demand systems. I first show that an individual firm exhibits pass-through in levels of cost shocks when its residual demand curve is log-linear, as under logit demand with atomistic firms. I then describe a broader class of models that predict complete pass-through in levels of common cost shocks.

EXAMPLE 3 (LOG-LINEAR DEMAND CURVES). Suppose the demand for good j can be written as

$$D_j\left(p_j, \boldsymbol{p_0}, Y\right) = \exp\left(a_j(\boldsymbol{p_0}, Y) - b_j(\boldsymbol{p_0}, Y)p_j\right).$$

Then $D_j\left(p_j, \boldsymbol{p_0}, Y\right)$ is shift-invariant in p_j with $\psi_j(\boldsymbol{p_0}, Y, \lambda) = \frac{1}{\lambda}\left(\exp\left(-b_j(\boldsymbol{p_0}, Y)\lambda\right) - 1\right)$.

The log-linear functional form means that level changes in a firm's price scale its demand curve up or down by a multiplicative factor. Several previous studies characterizing the pass-through of idiosyncratic cost shocks based on the shape of residual demand curves note this special property of log-linear demand curves (e.g., Bulow and Pfleiderer 1983; Weyl and Fabinger 2013; Mrázová and

Neary 2017). In some of this previous work, this special property is expressed in terms of the super-elasticity of demand: under log-linear demand, the elasticity of demand is proportional to price, $-\frac{\partial \log D_j}{\partial \log p_j} = p_j b_j(\boldsymbol{p_0}, Y)$, so that the super-elasticity is equal to one.

A popular special case of Example 3 is logit demand with atomistic firms.

EXAMPLE 4 (GENERALIZED LOGIT DEMAND WITH ATOMISTIC FIRMS). Suppose there are a continuum of goods indexed by $j \in [1, J]$ and an outside good with price p_0, and that the demand for good $j \in [1, J]$ is given by

$$D_j(\boldsymbol{p}, p_0, Y) = \frac{\exp\left(a_j(p_0, Y) - b_j \frac{p_j}{p_0}\right)}{\int_1^J \exp\left(a_k(p_0, Y) - b_k \frac{p_k}{p_0}\right) dk}, \tag{14}$$

where $\boldsymbol{p}$ is a vector of prices of goods $j \in [1, J]$ and $b_j > 0$ for all j. Then, demand is shift invariant with respect to any vector of prices $\boldsymbol{p}^{\text{subset}} \subseteq \boldsymbol{p}$.

When firms are atomistic, the influence of firm j's own price on the denominator in equation (14) is vanishingly small, and hence logit demand becomes a special case of Example 3. The price set by firm j is equal to its marginal cost plus an additive markup priced relative to the outside good,

$$p_j = c_j + \frac{p_0}{b_j}.$$

Thus, logit demand with atomistic firms yields complete pass-through in levels of any cost shock to inside firms, whether that shock is idiosyncratic to firm j, affects a subset of firms, or affects all inside firms $[1, J]$.

Log-linear demand curves and logit demand are functional forms that yield complete pass-through in levels of idiosyncratic shocks. However, a much broader class of demand systems satisfies shift invariance with respect to a set of firms' prices and thus generates complete pass-through in levels of common cost shocks, as I show in Example 5.

EXAMPLE 5 (DISCRETE CHOICE WITH QUASI-LINEAR PREFERENCES). Suppose there is a unit mass of consumers indexed by $i \in [0, 1]$, each with income Y. There are J inside goods and a single outside good (the numeraire). Each consumer purchases one unit of one of the J inside goods and spends the rest of her income on the numeraire. Consumer i's utility maximization problem is:

$$U_i = \max_j \{u_{ij}\} \qquad \text{s.t.} \qquad \begin{cases} u_{ij} = \delta_{ij} + q_{i0} & \text{(Utility)} \\ p_j + p_0 q_{i0} = Y & \text{(Budget constraint)} \end{cases},$$

where p_0 is the price of the numeraire, q_{i0} are units purchased of the numeraire, and δ_{ij} are consumer-specific tastes for each inside good. Assume further that consumers almost surely do not face ties in utility between goods.[26]

Then the demand system $D(\boldsymbol{p}, p_0, Y)$ given by aggregating over all consumers, so that

$$D_j(\boldsymbol{p}, p_0, Y) = \int_0^1 1\{u_{ij} > u_{ik} \text{ for all } k \neq j\}\, di$$

is shift invariant in $\boldsymbol{p}$ with $\psi_j = 0$.

The class of discrete-choice demand system in Example 5 is a special case of shift invariance where a common cost shock leaves firms' demand curves entirely unchanged (i.e., $\psi_j = 0$). This class of demand systems is sometimes referred to as translation-invariant choice systems (McFadden 1981) or linear random utility models (Anderson, de Palma, and Thisse 1992). Anderson, de Palma, and Thisse (1992) show that this class nests several demand systems used in the industrial organization literature as special cases. For example, logit, nested logit (Verboven 1996), mixed logit (Nevo 2001), multinomial probit, competition on a line (Hotelling 1929), and competition on a unit circle (Salop 1979) can all be expressed in terms of this framework.[27] Thus, each demand system is shift invariant with respect to inside prices and generates complete pass-through in levels of common cost shocks.[28]

26. That is, for any two goods j, n where $j \neq n$, the set of consumers where $a\delta_{ij} - b\delta_{in} = c$ for any positive constants $a, b > 0$ and for any constant c is measure zero.

27. Barro (2024) characterizes firms' markups in a variant of the Salop (1979) model with an intensive-margin elasticity of demand. When the intensive-margin elasticity of demand is zero, the Barro (2024) model coincides with a special case of Example 5. For this case, Barro (2024) shows that firms have additive markups and exhibit complete pass-through in levels, consistent with Proposition 5.

28. The quasilinear preferences in Example 5 rule out income effects. To capture the effect of large purchases on the marginal utility of leftover income, studies sometimes posit other ways for income and prices to enter the indirect-utility function. For example, Berry, Levinsohn, and Pakes (1995) and Petrin (2002) assume that indirect utility depends on $\log(Y_i - p_j)$. Griffith, Nesheim, and O'Connell (2018), Miravete, Seim, and Thurk (2023), and Birchall, Mohapatra, and Verboven (forthcoming) explore other departures from quasilinearity in discrete-choice mod-

Note that this class of demand systems also predicts that quantity shares do not change in response to a common cost shock, consistent with my evidence on rice and flour products in Online Appendix Table A13.

VII.E. Relationship to Pass-Through of Idiosyncratic Shocks

Scale and shift invariance determine how firms pass through common cost shocks. These conditions differ from previous work that describes how firms pass through idiosyncratic cost shocks. For example, Bulow and Pfleiderer (1983), Mrázová and Neary (2017), and Matsuyama and Ushchev (2021) characterize the pass-through of idiosyncratic cost shocks in terms of the elasticity and curvature of firms' residual demand curves. The pass-through in levels of an idiosyncratic cost shock to firm j, holding all other firms' prices fixed, is

$$\left.\frac{\partial p_j}{\partial c_j}\right|_{dp_n=0 \text{ for } n\neq j} = \frac{\sigma_j}{\sigma_j + \varepsilon_{jj} - 1},$$

where $\sigma_j = -\frac{\partial \log D_j}{\partial \log p_j}$ is the demand elasticity and $\varepsilon_{jj} = \frac{\partial \log \sigma_j}{\partial \log p_j}$ is the superelasticity of the firm's residual demand curve. Holding other prices fixed, firm j exhibits complete pass-through in levels of the idiosyncratic cost shock if the superelasticity of demand is equal to one, $\varepsilon_{jj} = 1$. (In equilibrium, even if $\varepsilon_{jj} = 1$, firm j's pass-through of an idiosyncratic cost shock may still differ from one if it internalizes how changing its price will affect its competitors' prices.)

The relevant statistic for the pass-through of common cost shocks is different. In the case where firms are identical and demand is symmetric, I can express the pass-through in levels of common cost shocks as

$$\frac{dp}{dc} = \frac{\sigma_j}{\sigma_j + \bar{\varepsilon}_j - 1}, \qquad \text{where} \qquad \bar{\varepsilon}_j = \varepsilon_{jj} + \sum_{n\neq j}^{J} \varepsilon_{jn},$$

and $\varepsilon_{jn} = \frac{\partial \log \sigma_j}{\partial \log p_n}$ reflects how firm j's demand elasticity changes with firm n's price. The statistic governing the pass-through of the common cost shock is the aggregate superelasticity $\bar{\varepsilon}_j$, which generally differs from the superelasticity of the firm's residual

els. In general, the discrete-choice demand systems they consider no longer satisfy shift invariance when prices enter the indirect-utility function nonlinearly.

demand curve.[29] (Examples 3 and 4 are special cases in which $\sum_{n\neq j}^{J}\varepsilon_{jn}=0$, so $\varepsilon_{jj}=\bar{\varepsilon}_j$.)

The gap between the aggregate superelasticity and the superelasticity of a firm's residual demand curve has two implications. First, the rate at which a firm passes through idiosyncratic cost shocks can generally differ from the rate at which it passes through common cost shocks. Imposing shift invariance disciplines the pass-through of common cost shocks while retaining flexibility to match different (and potentially firm-specific) pass-through rates for idiosyncratic cost shocks.

Second, two models can share identical residual demand superelasticities but have different predictions for the pass-through of common cost shocks. In their study of the coffee market, Nakamura and Zerom (2010) estimate a median superelasticity of residual demand of 4.64 and argue that this curvature helps their model generate aggregate, incomplete log pass-through. However, a Kimball demand system calibrated to match the same superelasticity of residual demand would instead predict complete log pass-through of common shocks. Thus, the superelasticity of residual demand curves is generally not sufficient to predict the pass-through of common cost shocks.

VIII. Quantitative Application

Here I illustrate how integrating shift-invariant demand systems into an otherwise standard input-output model of the U.S. economy affects its predictions for the dynamics of consumer price inflation. I argue that the standard model with fixed percentage markups predicts inflation that is too volatile and too responsive to upstream shocks. Incorporating shift-invariant demand—and therefore complete pass-through in levels—into the model improves its predictions for both the unconditional volatility of consumer price inflation and the response of inflation to identified shocks, while allowing for substantial markups in line with the microeconomic evidence.

29. Online Appendix B.1 shows that $\bar{\varepsilon}_j=1$ iff $\frac{\partial\psi_j(\boldsymbol{p},\boldsymbol{p_0},Y,0)}{\partial p_j}=0$, given the definition of ψ_j in Definition 2.

VIII.A. Setup

The economy consists of production labor and N goods that are used for consumption or as intermediate inputs in production. I take the wage for production labor and prices for a subset of goods $\mathcal{N}^{\text{exog}} \subset \{1, ..., N\}$ as exogenous. Each remaining good is produced by a single industry (I use subscripts i to refer to goods and industries interchangeably) that consists of a unit mass of firms indexed by f. Firms in each industry possess identical production functions, and the cost of producing y units for a firm in industry i is

$$C_i(y) = mc_i\,(p_1, ..., p_N, w)y + wF_i,$$

where w is the wage rate for labor, p_j is the price of good j, $mc_i(\cdot)$ is the marginal cost of production, and F_i is an overhead cost paid in units of labor.

The output of industry i is an aggregate of firms' differentiated varieties in the industry. I compare two cases. The first case assumes that each industry's output is a CES aggregate of firm varieties with an elasticity of substitution σ_i (allowed to vary across industries). This case implies that demand curves facing firms are scale invariant. In the absence of nominal rigidities, firms' desired prices equal a fixed percentage markup times marginal cost,

$$p^*_{if} = \frac{\sigma_i}{\sigma_i - 1} mc_i.$$

The second case assumes that the demand for the variety produced by firm f is given by the logit demand system

$$y_{if} = \frac{\exp\left(-b_i\,\frac{p_{if}}{w}\right)}{\int_0^1 \exp\left(-b_i\,\frac{p_{ig}}{w}\right)\,dg} y_i, \tag{15}$$

where y_i is total industry output. In the absence of nominal rigidities, these demand curves imply that firms' desired prices are equal to marginal cost plus an additive markup that is priced relative to the wage,[30]

$$p^*_{if} = mc_i + \frac{w}{b_i}.$$

The logit demand in equation (15) is a tractable special case for my purposes because, like CES demand, it implies that firms' desired prices depend only on their own marginal costs and not the marginal costs or prices of other firms.

I model nominal rigidities as Calvo (1983) frictions. In each period, only a fraction of firms δ_i in each sector are able to change their prices. In the presence of these nominal rigidities, the optimal reset price for firm f in industry i at time t solves

$$(16) \qquad p_{ift} = \arg\max_p \sum_{k=0}^{\infty} \beta^k (1-\delta_i)^k (p - mc_{it+k}) y_{ift+k}(p),$$

where $y_{ift+k}(p)$ is the demand for the firm at time $t+k$ with output price p, taking the prices set by all other firms as given.

Finally, I include a retail sector that produces a consumption good using a constant-returns production function over the N goods. I define the consumer price index as the price of a unit of output from the retail sector.

VIII.B. Calibration

I calibrate a log-linearized version of the model. Below, I briefly describe the data used for calibration. Online Appendix D provides a detailed account of the data sources, cleaning procedures, and solution method.

I take the 402 industries in the Bureau of Economic Analysis's (BEA) detailed input-output tables as the set of industries. I define $\mathcal{N}^{\text{exog}}$ as the set of Stage 1 industries designated by the BLS, which correspond to industries that are most upstream from consumer demand. I use monthly price indices for these industries from 1982–2018 from the BLS Producer Price Index

30. This pricing equation implies that changes in marginal cost arising from changes in production wages will be passed-through to prices more than one for one in levels, because wage changes also affect firms' desired additive markups. This relates to an observation by Okun (1981, 164) that firms appear to pass through changes in material costs to prices "essentially on a dollars-and-cents basis, while [changes in labor costs] are passed through with a percentage markup." In Online Appendix C.2, I provide empirical evidence of Okun's claim and show that the differential pass-through of labor costs relative to other inputs disappears once I account for the fact that additive markups are priced relative to the wage, as implied by equation (15).

(PPI) program, and I take the average hourly earnings of production and nonsupervisory employees as the measure of production wages.

Prices for the remaining industries are determined by firms' endogenous pricing decisions in the model. Equation (16) shows that these pricing decisions in each industry depend on future marginal costs, Calvo rigidities, and firms' desired markups. Marginal costs depend on the prices of goods that the industry uses as inputs; to a first order, the log deviation in industry i's marginal costs relative to steady state at time t is given by

$$d \log mc_{it} = \sum_{j} \Omega_{ij} d \log p_{jt}, \tag{17}$$

where Ω_{ij} is the steady-state share of sector i's variable costs spent on good j. I take these expenditure shares from the 2012 BEA tables.[31] For Calvo frictions in each industry, I use data on monthly frequencies of price adjustment by industry from Pasten, Schoenle, and Weber (2020), who compute these probabilities of price adjustment using the firm-level data that underlie the BLS PPI.[32] I set the monthly discount factor to $\beta = (0.96)^{\frac{1}{12}}$.

Finally, I calibrate firms' desired markups in each industry to match the industry's ratio of sales to variable costs. For the model with percentage markups, I consider three definitions of variable costs: (i) materials costs, (ii) materials costs and employee compensation, and (iii) materials costs, employee compensation, and consumption of fixed capital. For each definition, I set σ_i to match each industry's ratio of sales to variable costs. For the model with additive markups, I likewise choose b_i in each industry to match value added as a share of sales.

VIII.C. Results for Inflation Volatility

Table IX reports the volatility of consumer price inflation in the calibrated model from 1982 to 2018. The standard deviation of annual inflation rates in the model with fixed percentage markups, using material costs as the measure of industry's variable costs, is 2.5%, nearly double the volatility of consumer

31. Around the steady state, equation (17) describes changes in marginal costs up to a first order regardless of elasticities of substitution in production. In Online Appendix Table D1, I report results allowing for the expenditure shares Ω_{ij} to respond endogenously to prices due to input complementarity.

32. I am grateful to Raphael Schoenle for sharing these data.

TABLE IX
VOLATILITY OF CONSUMER PRICE INFLATION IN CALIBRATED INPUT-OUTPUT MODELS

	Std. dev. of annual inflation, 1982–2018 (%)	Cost-weighted average markup	
		Mfg.	All
Data			
PCE Price Index	1.1		
CPI for all urban consumers	1.3		
Model			
CES demand (percentage markups)			
Variable costs (VC) = materials	2.5	1.48	2.19
VC = materials + wages	1.7	1.19	1.31
VC = materials + wages + consumption of fixed capital	1.6	1.11	1.17
Logit demand (additive markups)	1.3	1.0–1.5	1.0–2.2

Notes. The cost-weighted average markup is the ratio of total industry sales to total industry variable costs, for manufacturing industries (Mfg.) and for all industries (All).

price inflation in the data over the same period. In other words, given the input-output structure of the economy and the volatility of commodity prices, the model with percentage markups generates too much volatility in consumer price inflation relative to the data.[33]

I can reduce the volatility of inflation predicted by the percentage markup model by expanding the definition of variable costs. Including wages and all other employee compensation costs in variable costs reduces the volatility of inflation rates in the model to 1.7% (still 30%–50% higher than the data). Further including consumption of fixed capital reduces the volatility of inflation to 1.6% (20%–40% higher than the data).

However, reducing the volatility of consumer price inflation in this way implies average markups that appear too low relative to markups estimated in microdata. The cost-weighted average markup across industries falls from 2.19 to 1.31 when I include labor costs, and falls further to 1.17 when I include consumption of fixed capital as part of variable costs. The implied markups for manufacturing industries are even lower. In comparison, De

33. If I assumed the central bank targets a desired volatility of consumer price inflation, the percentage-markups model predicts too little volatility in upstream commodity prices relative to the data.

Loecker, Eeckhout, and Unger (2020) estimate a cost-weighted average markup of 1.25 among public firms and average markups in excess of 1.7 for firms in the Census of Manufacturing. Estimates of this magnitude are also typical in industry-specific studies in the industrial organization literature.[34]

The model with additive markups reconciles this tension between the low volatility of consumer price inflation and plausibly sized markups. The volatility of consumer price inflation in the additive-markup model is 1.3%, in line with the volatility of consumer price index inflation and modestly larger than the volatility of personal consumption expenditures inflation. Moreover, the model can accommodate a cost-weighted average markup anywhere between 1.0 and 2.2. Although larger markups in the percentage markup model amplify the effect of upstream price movements on downstream prices and inflation, in the additive markup model, upstream price movements are passed through in levels regardless of the size of markups.

VIII.D. Results for an Oil Price Shock

The input-output model with additive markups not only accounts for the low volatility of consumer price inflation but also better matches the response of inflation to identified shocks. I demonstrate this using shocks to oil production constructed by Känzig (2021). The left panel of Figure V shows the path of oil prices following an oil supply news shock, which I feed into the calibrated model.[35]

The right panel of Figure V compares the response of consumer price inflation in the percentage markup and additive

34. The relevant statistic for pass-through in the percentage-markup model is the ratio of sales to variable costs. Studies that find evidence of lower aggregate markups are typically referring to markups as the ratio of sales to the sum of variable and overhead costs: for example, Gutiérrez and Philippon (2017), who find an aggregate markup around 1.10, estimate markups as firms' operating margins less depreciation.

35. I assume that the shock to the oil price, measured using the West Texas Intermediate crude oil price by Känzig (2021), corresponds to a shock to the Oil and Gas Extraction (211000) price in the model. Despite the Oil and Gas Extraction industry covering a broader scope of products, the two price series in the data are strongly correlated and exhibit similar volatilities. Over the period 1982–2018, the Oil and Gas Extraction industry price index and the West Texas Intermediate crude price have a correlation of 0.95, and the volatility of inflation rates for the Oil and Gas Extraction industry price index is 30%, compared with 32% for the West Texas Intermediate crude price.

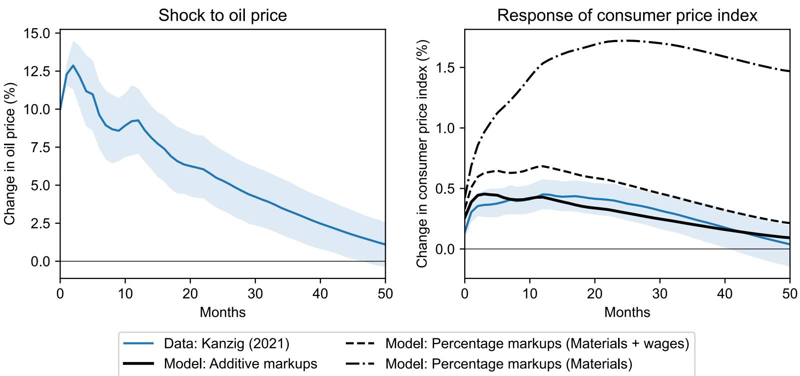


FIGURE V

Response of the Consumer Price Index to Oil Supply Shocks in Data and Model

The thin solid blue lines are impulse responses to an oil supply news shock from Känzig (2021) (shaded areas indicate 68% confidence bands). Changes in the oil price in the left panel are treated as exogenous changes in the oil price in the model. In the right panel, the black lines plot the response of the personal consumption expenditures price index to the oil price shock in the calibrated models.

markup models to the data. Känzig (2021) estimates that the consumer price index reaches a maximum response of 45 log points at 12 months after an oil supply news shock. Compared with the data, the percentage-markup model predicts too much responsiveness in consumer price inflation: at 12 months the predicted response of the consumer price index is 56 log points (153 log points) when assuming variable costs include material and labor costs (material costs alone). In comparison, the response of consumer price inflation in the additive markup model lines up remarkably well with the data. The peak response at 12 months is 43 log points, and the predicted response of the consumer price index to the oil shock falls within the 68% confidence interval from Känzig (2021) at all horizons. Thus, the additive markup model is better able to match the data than the model with percentage markups. As stressed, the additive markups model also achieves this responsiveness while allowing for substantial firm markups consistent with micro evidence.

IX. CONCLUSION

Incomplete log pass-through and adjustment in firms' percentage markups may be better understood in terms of complete pass-through in levels and fixed additive markups. Across a broad array of markets, I find that complete pass-through in levels explains both the extent of incomplete log pass-through and cross-sectional variation in log pass-through. Complete pass-through in levels may help explain several other features of the data, such as asymmetry and size dependence in log pass-through, heterogeneity in log pass-through by firm size and product quality, changes in relative price and markup dispersion, and dynamics of industry profits and entry.

Under standard assumptions about firm conduct, the homothetic demand systems commonly used in macroeconomics and trade are incompatible with my evidence. I propose that an alternative class of shift-invariant demand systems can generate complete pass-through in levels of common cost shocks while maintaining standard conduct assumptions. I illustrate how to integrate these demand systems into an input-output model of the U.S. economy. I anticipate that integrating demand systems consistent with pass-through in levels into other workhorse models may be useful for understanding other features of the data.

Supplementary Material

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

Data Availability

The data and code underlying this article is available in Sangani (2026), in the Harvard Dataverse, https://doi.org/10.7910/DVN/SNJDM4.

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